

Dynamic properties of an SARS-CoV-2 epidemic model via stochastic PINNs

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ABSTRACT

This paper introduces a novel stochastic SEIRV model for investigating the spread of the SARS-CoV-2 epidemic using stochastic physics-informed neural networks (S-PINNs). We first prove the global positivity of solutions via Lyapunov functions and Itô's formula. Then, persistence and extinction properties are analyzed by a threshold. The S-PINNs algorithm integrates noise into the loss function for seamless data-driven and physics-based learning. Finally, the algorithm based on the SEIRV model is applied to SARS-CoV-2 data from Austria, Switzerland, and Belgium, and it outperforms PINNs, LSTMs, and Logistic models, especially on noisy data.

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1. Introduction

Infectious diseases continue to pose serious threats to human health and socio-economic stability in the context of globalization and high population mobility. During the SARS-CoV-2 pandemic, the world experienced significant excess mortality (Almeida et al., 2024; Wakefield et al., 2024; Wang et al., 2022), and life expectancy declined in 29 countries in 2020 (Aburto et al., 2022), with some countries experiencing decreases not seen since World War II (Islam et al., 2021). Studies have shown that SARS-CoV-2 not only caused direct deaths but also produced indirect harm through delayed treatment of chronic diseases, healthcare resource shortages, and economic stagnation (Barbiellini Amidei et al., 2025; Huang et al., 2024). In addition, long-term sequelae ("Long COVID") have had profound impacts on individual health, labor markets, and healthcare systems (Davis et al., 2021). These results indicate that the harms of infectious diseases are long-term and systemic, going far beyond infections and mortality alone (Karlinsky & Kobak, 2022).

In mathematical epidemiology, the classical SIR model, proposed by Kermack and McKendrick, provides a foundational framework to understand epidemic thresholds and wave dynamics (Kermack & McKendrick, 1927). Subsequently, an exposed compartment (E) was introduced to form the SEIR model, which is better suited to describing diseases with incubation periods, such as SARS-CoV-2. With vaccination becoming a key measure for epidemic control, the SEIRV model was further developed (Gupta et al., 2024). Recent studies have demonstrated that SEIRV and its extensions can be applied to assess the effects of different vaccination coverages, waning immunity, reinfection mechanisms, and heterogeneous contact patterns on epidemic spread (Ali et al., 2025; Tridane et al., 2023). For example, SEIRV frameworks have been used to analyze

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multivaccine strategies and vaccination sequences to optimize epidemic control (Santos Junior & Gondim, 2024). Other studies have investigated the roles of waning immunity and age structure in recurrent epidemic waves (Garnier et al., 2024; Gomes et al., 2024). These studies demonstrate that SEIRV models play a critical role in bridging theoretical modeling with public health policy.

Taking into account the dissemination of information during a disease outbreak, various social media platforms utilize diverse channels to distribute details regarding symptoms, preventive strategies, and therapeutic options, while also enhancing awareness among governmental bodies and the general populace through educational initiatives. The government enacts regulatory measures proportional to the severity of the epidemic, recommending that individuals adopt appropriate protective measures. Hence, governmental policies and societal responses are pivotal in mitigating disease propagation (Saha et al., 2022). Let the sociological parameter η represent the intensity of government intervention, and $\mathcal{D}$ and $\mathcal{K}$ denote the dynamic changes in social behavior and the degree of public response, respectively. Under reasonable assumptions that $0 \leq \eta < 1$ and $0 \leq \mathcal{D} < 1$, the transmission rate can be expressed as $(1 - \eta)(1 - \mathcal{D})^{\mathcal{K}}\beta$, which describes the process by which susceptible individuals transition to the infected state. Incorporating these factors, we establish the following deterministic SEIRV model:

$$\begin{cases} dS(t) = [\Lambda - (1 - \eta)(1 - \mathcal{D})^{\mathcal{K}}\beta S(t)I(t) - \mu_S S(t) - \epsilon S(t)] dt, \\ dE(t) = [(1 - \eta)(1 - \mathcal{D})^{\mathcal{K}}\beta S(t)I(t) - \alpha_E E(t) - \mu_E E(t)] dt, \\ dI(t) = [\alpha_E E(t) - \mu_I I(t) - \gamma_I I(t)] dt, \\ dR(t) = [\gamma_I I(t) - \mu_R R(t)] dt, \\ dV(t) = [\epsilon S(t) - \mu_V V(t)] dt, \end{cases} \quad (1)$$

where the constant Λ represents the recruitment or birth rate into the susceptible, accounting for population influx. The transmission rate β quantifies the probability of disease transmission from infected to susceptible individuals upon contact, typically under the bilinear incidence assumption. The natural mortality rates for each compartment are specified as μ_S , μ_E , μ_I , μ_R , and μ_V , reflecting demographic losses independent of the disease. Additionally, α_E is the rate at which exposed individuals progress to the infectious state. The parameter γ_I is the recovery rate for infected individuals, encompassing both natural immunity acquisition and medical interventions. ϵ denotes the vaccination rate, representing the proportion of susceptible individuals transitioning to the vaccinated compartment per unit time, which is crucial for assessing control strategies. All parameters are positive constants. Since the R compartment does not directly participate in disease transmission and our research focus is on the control effects of vaccination, we omit it. This simplification reduces model complexity without affecting the core analysis of transmission dynamics. Based on this, the model (1) is rewritten by

$$\begin{cases} dS(t) = [\Lambda - (1 - \eta)(1 - \mathcal{D})^{\mathcal{K}}\beta S(t)I(t) - \mu_S S(t) - \epsilon S(t)] dt, \\ dE(t) = [(1 - \eta)(1 - \mathcal{D})^{\mathcal{K}}\beta S(t)I(t) - \alpha_E E(t) - \mu_E E(t)] dt, \\ dI(t) = [\alpha_E E(t) - \mu_I I(t) - \gamma_I I(t)] dt, \\ dV(t) = [\epsilon S(t) - \mu_V V(t)] dt. \end{cases} \quad (2)$$

As shown in Fig. 1, we can observe the personnel movement between the different compartments. In reality, the spread of infectious diseases is subject to stochasticity, including fluctuations in contact patterns, uncertainty in transmission rates, and variability in the implementation of interventions. Consequently, researchers have incorporated stochastic

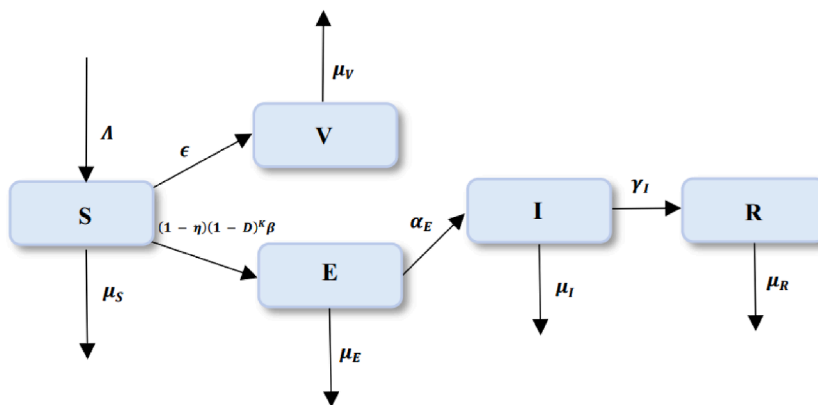


Fig. 1. The SEIRV epidemic model describes the transitions among five compartments Susceptible (S), Exposed (E), Infectious (I), Recovered (R), and Vaccinated (V) with the transmission rate β and other epidemiological parameters.

perturbations into SEIR/SEIRV models to investigate extinction probabilities, stationary distributions, and threshold corrections (Jiang et al., 2023; Wang et al., 2021; Zhao et al., 2024). Environmental stochastic variations, including temperature and weather fluctuations, cause model parameters to vary unpredictably. From a biological standpoint, these uncontrollable, random environmental factors significantly alter infectious disease transmission trajectories, as exemplified by the influenza virus exhibiting enhanced transmissibility during declining temperatures, thereby modifying conventional disease propagation patterns. Consequently, infectious disease models incorporating environmental noise perturbations provide more accurate representations of disease dynamics than deterministic models. We posit that environmental noise critically affects each state in the model (2), assuming that each population group, as a cohesive unit, is collectively influenced by these environmental disturbances (Khan et al., 2019, 2021; Wang et al., 2023). Building upon model (2), we conduct a detailed investigation of a stochastic SEIRV framework that integrates the effects of governmental interventions and public responses as follows

$$\begin{cases} dS(t) = [\Lambda - (1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \mu_S S(t) - \epsilon S(t)]dt + \sigma_1 S(t)dB_1(t), \\ dE(t) = [(1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \alpha_E E(t) - \mu_E E(t)]dt + \sigma_2 E(t)dB_2(t), \\ dI(t) = [\alpha_E E(t) - \mu_I I(t) - \gamma_I I(t)]dt + \sigma_3 I(t)dB_3(t), \\ dV(t) = [\epsilon S(t) - \mu_V V(t)]dt + \sigma_4 V(t)dB_4(t). \end{cases} \quad (3)$$

The stochastic model is formulated within a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where $B_i(t) (i = 1, 2, 3, 4)$ denote independent standard Brownian motions, originating at $t = 0$. The coefficients $\sigma_i(t) (i = 1, 2, 3, 4)$ represent positive constants that measure the magnitude of stochastic noise, effectively capturing the variability induced by perturbations. A notable attribute of the stochastic model (3) is its innovative incorporation of two fundamental biological mechanisms: the virus's latency phase and the logistic growth of susceptible populations, while assuming that the reduction in fertility among infected individuals can be overlooked.

Physics-Informed Neural Networks (PINNs) have been widely used in SARS-CoV-2 modeling by embedding epidemiological dynamics into neural networks (Galasso et al., 2024; Millevoi et al., 2024; Xu et al., 2023). Recent studies highlight their strengths in parameter estimation, state reconstruction, multi-wave forecasting, and intervention optimization, particularly under time-varying transmission, waning immunity, and complex controls. For example, Millevoi et al. (Millevoi et al., 2024) inferred time-varying SIR parameters from early Italian COVID-19 data in low-data regimes. Han et al. (Han et al., 2024) captured Germany's epidemic dynamics and trends via SIR-PINNs. Qian et al. (Qian et al., 2025) achieved superior 1–4 week forecasts for California using cases, hospitalizations, deaths, vaccination, and mobility data, outperforming data-driven models when modeling time-varying effects. Cheng et al. (Cheng et al., 2025) enhanced SEIRV temporal predictions with DINNs, showing robust performance under noise. Stochastic extensions, such as SPINODE (O'Leary et al., 2022) and physics-informed GANs (Yang et al., 2020), further enable uncertainty-aware modeling of noisy epidemic systems. These methods excel in data-scarce, non-stationary, and multi-intervention settings by tightly fusing mechanistic knowledge with observations, improving interpretability and reliability. Therefore, stochastic epidemiological models based on advanced PINNs should be considered to offer powerful tools for epidemic modeling and public health decision-making.

Based on the above analysis, we focus on the dynamic properties of the stochastic model (3) via a stochastic PINN. The rest of this paper is organized as follows. In Section 2, we investigate the existence and uniqueness, ergodicity, and extinction properties of the stochastic model (3), with the main theoretical results presented in Theorems 1, 2, and 3. To enhance comprehension, we conduct numerical simulations in Section 3 to validate the theoretical findings presented earlier. Our main results demonstrate that model (3) exhibits excellent fitting properties and biological significance. Furthermore, in Section 4, we introduce the Stochastic PINNs (S-PINNs) algorithm and integrate the stochastic SEIRV model. In Section 5, based on SARS-CoV-2 data from Austria, we compare the S-PINNs algorithm with PINNs, LSTMs, and logistic models to demonstrate its effectiveness in handling noisy data. A brief conclusion is provided in Section 6.

2. Dynamic properties

Based on the stochastic SEIRV model (3), we systematically investigate the dynamical properties of the model in this section. The existence and uniqueness of positive solutions for the stochastic system ensure mathematical rigor and biological interpretability. The ergodic properties of the model focus on the existence of stationary distributions and the long-term behavior. Finally, we study the conditions for disease extinction, determining the parameter ranges for which stochastic perturbations lead to disease elimination. These theoretical analyses provide a mathematical foundation for understanding the impact of environmental noise on disease transmission, offering a basis for formulating effective control strategies.

2.1. The positive solution

Denote $\mathbb{R}_+ = [0, \infty)$, $\mathbb{R}_{++} = (0, \infty)$. Let $Q_0 = (S(0), E(0), I(0), V(0))$ be the starting value, and $Q_t = (S(t), E(t), I(t), V(t))|_{t \geq 0}$ is the solution of model (3). Note that, as a consequence of Itô's formula, the infinitesimal generator of model (3) is the following second-order differential operator: for each $(S, E, I, V) \in \mathbb{R}^4$, we have

$$\begin{aligned} \mathcal{L} = & \left[\Lambda - (1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \mu_S S(t) - \epsilon S(t) \right] \frac{\partial}{\partial S} \\ & + \left[(1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \alpha_E E(t) - \mu_E E(t) \right] \frac{\partial}{\partial E} \\ & + [\alpha_E E(t) - \mu_I I(t) - \gamma_I I(t)] \frac{\partial}{\partial I} + [\epsilon S(t) - \mu_V V(t)] \frac{\partial}{\partial V} \\ & + \frac{1}{2} \sigma_1^2 S^2 \frac{\partial^2}{\partial S^2} + \frac{1}{2} \sigma_2^2 E^2 \frac{\partial^2}{\partial E^2} + \frac{1}{2} \sigma_3^2 I^2 \frac{\partial^2}{\partial I^2} + \frac{1}{2} \sigma_4^2 V^2 \frac{\partial^2}{\partial V^2}. \end{aligned} \quad (4)$$

Theorem 1. For any initial value $Q_0 \in \mathbb{R}_+^4$, the stochastic model (3) possesses a unique global continuous solution Q_t valued in $\mathbb{R}_+^4$.

Proof. For the initial value $Q_0 \in \mathbb{R}_+^4$, there exists a unique local solution $Q_t \in \mathbb{R}_+^4$ to model (3) defined on the interval $t \in [0, \tau)$, where the explosion time τ is specified as

$$\tau = \inf \left\{ t \in [0, \infty) : Q_t \notin \mathbb{R}_+^4 \text{ or } |S(t)| + |E(t)| + |I(t)| + |V(t)| = \infty \right\}.$$

This follows from the local Lipschitz continuity of the coefficients of the stochastic model. Our goal is to prove that $\tau = \infty$ almost surely, implying that model (3) admits a unique global solution Q_t valued in $\mathbb{R}_+^4$. To achieve this, select a sufficiently large $y_0 > 1$ such that the interval $[1/y_0, y_0]$ encompasses all components of Q_0 . For $y > y_0$, define

$$\tau_y = \inf \left\{ t \in [0, \tau) : \max\{S(t), E(t), I(t), V(t)\} \geq y \text{ or } \min\{S(t), E(t), I(t), V(t)\} \leq \frac{1}{y} \right\} = \inf D.$$

Observe that if $D = \emptyset$, then $\tau = \infty$, and consequently $\tau_y = \inf \emptyset = \tau = \infty$. Clearly, τ_y is non-decreasing in y . Set $\tau_\infty = \lim_{y \rightarrow \infty} \tau_y$. It holds that $\tau_\infty \leq \tau$ almost surely. Thus, establishing $\tau_\infty = \infty$ almost surely will suffice to show $\tau = \infty$ almost surely. Assume, for contradiction, that $\mathbb{P}\{\tau_\infty < \infty\} > 0$. Then, there exist constants $\zeta \in (0, 1)$, $T \in (0, \infty)$, and $y_1 \in (y_0, \infty)$ such that $\mathbb{P}\{\tau_\infty \leq T\} > \zeta$ and $\mathbb{P}\{\tau_y \leq T\} \geq \zeta$ for all $y \geq y_1$. Construct a C^2 -function $V : \mathbb{R}_+^4 \rightarrow \mathbb{R}_+$ given by

$$\mathbb{V}(S, E, I, V) = \left(S - c - c \ln \frac{S}{c} \right) + (E - 1 - \ln E) + (I - 1 - \ln I) + (V - 1 - \ln V),$$

where c is a constant to be determined later, and the non-negativity of this function stems from the inequality $u - 1 - \ln u \geq 0$ for $u > 0$. Define

$$\mathbb{V}(t) = \mathbb{V}(S(t), E(t), I(t), V(t)) \quad \text{for } t \geq 0.$$

Using Itô's formula, we can obtain:

$$\begin{aligned} d\mathbb{V}(t) = & \left\{ \left(1 - \frac{c}{S(t)} \right) \left[\Lambda - (1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \mu_S S(t) - \epsilon S(t) \right] + \frac{1}{2} \sigma_1^2 c \right\} dt \\ & + \left\{ \left(1 - \frac{1}{E(t)} \right) \left[(1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \alpha_E E(t) - \mu_E E(t) \right] + \frac{1}{2} \sigma_2^2 \right\} dt \\ & + \left\{ \left(1 - \frac{1}{I(t)} \right) [\alpha_E E(t) - \mu_I I(t) - \gamma_I I(t)] + \frac{1}{2} \sigma_3^2 \right\} dt + \left(1 - \frac{1}{I(t)} \right) \sigma_3 I(t) dB_3(t) \\ & + \left\{ \left(1 - \frac{1}{V(t)} \right) [\epsilon S(t) - \mu_V V(t)] + \frac{1}{2} \sigma_4^2 \right\} dt + \left(1 - \frac{1}{V(t)} \right) \sigma_4 V(t) dB_4(t) \\ & + \left(1 - \frac{c}{S(t)} \right) \sigma_1 S(t) dB_1(t) + \left(1 - \frac{1}{E(t)} \right) \sigma_2 E(t) dB_2(t) \\ = & (\mathcal{L}\mathbb{V})(t) dt + (S(t) - c) \sigma_1 dB_1(t) + (E(t) - 1) \sigma_2 dB_2(t) + (I(t) - 1) \sigma_3 dB_3(t) + (V(t) - 1) \sigma_4 dB_4(t), \end{aligned}$$

where

$$\begin{aligned}
(\mathcal{L}\mathbb{V})(t) &= \Lambda - (1-\eta)(1-\mathcal{D})^\kappa \beta S(t)I(t) - \mu_S S(t) - \epsilon S(t) + (1-\eta)(1-\mathcal{D})^\kappa \beta S(t)I(t) \\
&\quad - \alpha_E E(t) - \mu_E E(t) + \alpha_E E(t) - \mu_I I(t) - \gamma_I I(t) + \epsilon S(t) - \mu_V V(t) - \frac{c\Lambda}{S(t)} \\
&\quad + c(1-\eta)(1-\mathcal{D})^\kappa \beta I(t) + c\mu_S + c\epsilon - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta S(t)I(t)}{E(t)} + \alpha_E + \mu_E \\
&\quad - \frac{\alpha_E E(t)}{I(t)} + \mu_I + \gamma_I - \frac{\epsilon S(t)}{V(t)} + \mu_V + \frac{1}{2} (c\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2) \\
&\leq \left[c(1-\eta)(1-\mathcal{D})^\kappa \beta - (\mu_I + \gamma_I) \right] I(t) + \Lambda + c\mu_S + c\epsilon + \alpha_E + \mu_E + \mu_I + \gamma_I + \mu_V \\
&\quad + \frac{1}{2} (c\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2).
\end{aligned}$$

Denote $c = \frac{\mu_I + \gamma_I}{(1-\eta)(1-\mathcal{D})^\kappa \beta}$ and define $A_1 := \Lambda + c\mu_S + c\epsilon + \alpha_E + \mu_E + \mu_I + \gamma_I + \mu_V + \frac{1}{2} (c\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2)$. Consequently, $(\mathcal{L}\mathbb{V})(t) \leq A_1$, where A_1 is a positive constant. Therefore,

$$\begin{aligned}
\int_0^{\tau_y \wedge T} d\mathbb{V}(t) &\leq \int_0^{\tau_y \wedge T} A_1 dt + \int_0^{\tau_y \wedge T} (S(t) - c) \sigma_1 dB_1(t) + \int_0^{\tau_y \wedge T} (E(t) - 1) \sigma_2 dB_2(t) \\
&\quad + \int_0^{\tau_y \wedge T} (I(t) - 1) \sigma_3 dB_3(t) + \int_0^{\tau_y \wedge T} (V(t) - 1) \sigma_4 dB_4(t).
\end{aligned}$$

Taking the expectation yields

$$\mathbb{E}[\mathbb{V}(\tau_y \wedge T)] \leq \mathbb{V}(Q_0) + \mathbb{E} \left[\int_0^{\tau_y \wedge T} A_1 dt \right] \leq \mathbb{V}(Q_0) + A_1 T.$$

Define $\Omega_y = \{\tau_y \leq T\}$. As can be seen from the above, it follows that $\mathbb{P}(\Omega_y) \geq \eta$ for every $y \geq y_1$. Note that for any $a \in \Omega_y$, at least one of $S(\tau_y(a))$, $E(\tau_y(a))$, $I_1(\tau_y(a))$, or $I_2(\tau_y(a))$ equals either y or $1/y$. Thus,

$$\mathbb{V}(\tau_y(a)) \geq \min \left\{ y - 1 - \ln y, \frac{1}{y} - 1 - \ln \frac{1}{y}, y - c - \ln \frac{y}{c}, \frac{1}{y} - c - \ln \frac{1}{cy} \right\}.$$

It follows that

$$\begin{aligned}
\mathbb{V}(Q_0) + A_1 T &\geq \mathbb{E} \left[1_{\Omega_y} \mathbb{V}(\tau_y \wedge T) \right] \\
&\geq \eta \min \left\{ y - 1 - \ln y, \frac{1}{y} - 1 - \ln \frac{1}{y}, y - c - \ln \frac{y}{c}, \frac{1}{y} - c - \ln \frac{1}{cy} \right\},
\end{aligned}$$

where 1_{Ω_y} denotes the indicator function of Ω_y . As $y \rightarrow \infty$, $\mathbb{V}(Q_0) + A_1 T \geq \infty$, which contradicts the finiteness of $\mathbb{V}(Q_0) + A_1 T$. Hence, $\tau_\infty = \infty$ almost surely.

2.2. Ergodicity

In exploring epidemic models, we aim to gain a deeper understanding of the potential duration of disease persistence within biological populations. In this subsection, we leverage Khasminskii's theory (Khasminskii, 2012) to demonstrate that the stochastic SEIRV model possesses a unique ergodic stationary distribution. The existence of such a distribution implies that, in the long term, the disease will not vanish entirely but will persist and fluctuate probabilistically within the population, thereby reflecting the persistence and inherent uncertainty of real-world epidemics. To proceed, assume the Markov process $(X(t))_{t \geq 0}$ is time-homogeneous within the positive orthant $\mathbb{R}_+^4$, and we may formally analyze the associated stochastic differential equation:

$$dX(t) = b(X(t)) dt + \sum_{r=1}^k g_r(X(t)) dB_r(t),$$

where $b = (b^i)_{1 \leq i \leq 4} : \mathbb{R}_+^4 \rightarrow \mathbb{R}^4$ represents a measurable vector-valued function, the processes $B_r = (B_r(t))_{t \geq 0} (1 \leq r \leq k)$ are independent standard one-dimensional Brownian motions, and each $g_r = (g_r^i)_{1 \leq i \leq 4} : \mathbb{R}_+^4 \rightarrow \mathbb{R}^4$ is a measurable function. The diffusion matrix and the infinitesimal generator are defined as

$$A(x) = (a_{ij}(x))_{1 \leq i, j \leq 4}, \quad a_{ij}(x) = \sum_{r=1}^k g_r^i(x) g_r^j(x), \quad \mathcal{L} = \sum_{i=1}^4 b^i(x) \frac{\partial}{\partial x_i} + \frac{1}{2} \sum_{i, j=1}^4 a_{ij}(x) \frac{\partial^2}{\partial x_i \partial x_j}, \quad x \in \mathbb{R}_+^4.$$

Then, from Khasminskii (Khasminskii, 2012), the following result holds.

Lemma 1. Suppose there exists a bounded region $\mathcal{N} \subset \mathbb{R}_+^4$ possessing a smooth boundary and satisfying conditions: (i) There exists a positive constant $K > 0$ such that $\sum_{i, j=1}^4 a_{ij}(x) \xi_i \xi_j \geq K |\xi|^2$ holds for every $x \in \mathcal{N}$ and all $\xi = (\xi_1, \xi_2, \xi_3, \xi_4) \in \mathbb{R}^4$, and (ii) A non-negative C^2 -function $\mathcal{V} : \mathbb{R}_+^4 \rightarrow \mathbb{R}$ exists, along with a positive constant $L > 0$, satisfying $\mathcal{L}\mathcal{V}(x) \leq -L, x \in \mathbb{R}_+^4 \setminus \mathcal{N}$. Under these assumptions, the Markov process $(X(t))_{t \geq 0}$ admits a unique ergodic invariant measure.

Furthermore, for each initial state $x \in \mathbb{R}_+^4$ and every function $f \in L^1(\mathbb{R}_+^4, \mu)$,

$$\mathbb{P} \left\{ \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(X(t)) dt = \int_{\mathbb{R}_+^4} f(y) \mu(dy) \mid X(0) = x \right\} = 1.$$

Referring back to model (3), define a threshold value

$$R_0^S := \frac{(1 - \eta)(1 - \mathcal{D})^K \beta \alpha_E \epsilon}{\left(\mu_S + \epsilon + \frac{1}{2} \sigma_1^2 \right) \left(\alpha_E + \mu_E + \frac{1}{2} \sigma_2^2 \right) \left(\mu_I + \gamma_I + \frac{1}{2} \sigma_3^2 \right) \left(\mu_V + \frac{1}{2} \sigma_4^2 \right)}.$$

Theorem 2. If $R_0^S > 1$, then the model (3) possesses a unique stationary distribution and exhibits ergodicity for any initial value $Q_0 \in \mathbb{R}_+^4$.

Proof. The diffusion matrix $(a_{ij}(S, E, I, V))_{1 \leq i, j \leq 4}$ of the stochastic model (3) is given by

$$\begin{pmatrix} \sigma_1^2 S^2 & 0 & 0 & 0 \\ 0 & \sigma_2^2 E^2 & 0 & 0 \\ 0 & 0 & \sigma_3^2 I^2 & 0 \\ 0 & 0 & 0 & \sigma_4^2 V^2 \end{pmatrix}.$$

Let $M_0 := \min_{(S, E, I, V) \in \mathcal{N}_k \cap \mathbb{R}_+^4} \{ \sigma_1^2 S^2, \sigma_2^2 E^2, \sigma_3^2 I^2, \sigma_4^2 V^2 \}$, where for any $k > 1$,

$$\mathcal{N}_k := \left[\frac{1}{k}, k \right] \times \left[\frac{1}{k^3}, k^3 \right] \times \left[\frac{1}{k}, k \right] \times \left[\frac{1}{k}, k \right]. \quad (5)$$

Then, for all $(S, E, I, V) \in \mathcal{N}_k$ and $\eta = (\eta_1, \eta_2, \eta_3, \eta_4) \in \mathbb{R}^4$, the quadratic form

$$\sum_{i, j=1}^4 a_{ij}(S, E, I, V) \eta_i \eta_j = (\sigma_1 S)^2 \eta_1^2 + (\sigma_2 E)^2 \eta_2^2 + (\sigma_3 I)^2 \eta_3^2 + (\sigma_4 V)^2 \eta_4^2 \geq M_0 \|\eta\|^2.$$

Thus, Lemma 1 (i) holds. To establish Lemma 1 (ii), define a C^2 -function $U_1 : \mathbb{R}_+^4 \rightarrow \mathbb{R}$ by

$$U_1 = -\ln S - b_1 \ln E - b_2 \ln I - b_3 \ln V,$$

where b_1, b_2 , and b_3 are positive constants. Refer to Equation (4) for details, we obtain:

$$\begin{aligned}
\mathcal{L}U_1 &= -\frac{b_1(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{b_2\alpha_E E}{I} - \frac{b_3\epsilon S}{V} - \frac{V}{S^2} + \mu_S + \epsilon + b_1\alpha_E + b_1\mu_E \\
&\quad + b_2\mu_I + b_2\gamma_I + b_3\mu_V + (1-\eta)(1-\mathcal{D})^\kappa \beta I + \frac{V}{S^2} + \frac{1}{2}(\sigma_1^2 + b_1\sigma_2^2 + b_2\sigma_3^2 + b_3\sigma_4^2) \\
&= -\frac{b_1(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{b_2\alpha_E E}{I} - \frac{b_3\epsilon S}{V} - \frac{V}{S^2} + b_1\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right) \\
&\quad + b_2\left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right) + b_3\left(\mu_V + \frac{1}{2}\sigma_4^2\right) + (1-\eta)(1-\mathcal{D})^\kappa \beta I + \frac{V}{S^2} + \mu_S + \epsilon + \frac{1}{2}\sigma_1^2 \\
&\leq -4\left((1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon b_1 b_2 b_3\right)^{\frac{1}{4}} + b_1\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right) + b_2\left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right) \\
&\quad + (1-\eta)(1-\mathcal{D})^\kappa \beta I + b_3\left(\mu_V + \frac{1}{2}\sigma_4^2\right) + \frac{V}{S^2} + \mu_S + \epsilon + \frac{1}{2}\sigma_1^2.
\end{aligned}$$

Let

$$b_1\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right) = b_2\left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right) = b_3\left(\mu_V + \frac{1}{2}\sigma_4^2\right) = \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon}{\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right)\left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right)\left(\mu_V + \frac{1}{2}\sigma_4^2\right)}.$$

We can obtain:

$$\begin{aligned}
b_1 &= \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon}{\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right)^2 \left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right) \left(\mu_V + \frac{1}{2}\sigma_4^2\right)}, \\
b_2 &= \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon}{\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right) \left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right)^2 \left(\mu_V + \frac{1}{2}\sigma_4^2\right)}, \\
b_3 &= \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon}{\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right) \left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right) \left(\mu_V + \frac{1}{2}\sigma_4^2\right)^2}.
\end{aligned}$$

Thus,

$$\begin{aligned}
\mathcal{L}U_1 &\leq (1-\eta)(1-\mathcal{D})^\kappa \beta I + \frac{V}{S^2} + \mu_S + \epsilon + \frac{1}{2}\sigma_1^2 + 3 \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon}{\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right)\left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right)\left(\mu_V + \frac{1}{2}\sigma_4^2\right)} \\
&\quad + \left(\frac{\left((1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon\right)^4}{\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right)^4 \left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right)^4 \left(\mu_V + \frac{1}{2}\sigma_4^2\right)^4} \right)^{\frac{1}{4}} \\
&= \left(\mu_S + \epsilon + \frac{1}{2}\sigma_1^2\right) \left(- \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta \alpha_E \epsilon}{\left(\mu_S + \epsilon + \frac{1}{2}\sigma_1^2\right)\left(\alpha_E + \mu_E + \frac{1}{2}\sigma_2^2\right)\left(\mu_I + \gamma_I + \frac{1}{2}\sigma_3^2\right)\left(\mu_V + \frac{1}{2}\sigma_4^2\right)} + 1 \right) \\
&\quad + (1-\eta)(1-\mathcal{D})^\kappa \beta I + \frac{V}{S^2} \\
&= -\left(\mu_S + \epsilon + \frac{1}{2}\sigma_1^2\right)(R_0^S - 1) + (1-\eta)(1-\mathcal{D})^\kappa \beta I + \frac{V}{S^2} \\
&= -\lambda + (1-\eta)(1-\mathcal{D})^\kappa \beta I + \frac{V}{S^2},
\end{aligned}$$

where $\lambda = \left(\mu_S + \epsilon + \frac{1}{2}\sigma_1^2\right)(R_0^S - 1) > 0$.

Let $U_2 = -\ln E$, we can get

$$\mathcal{L}U_2 = -\frac{1}{E}((1-\eta)(1-\mathcal{D})^\kappa \beta SI - \alpha_E E - \mu_E E + \frac{\sigma_2^2}{2}) = -\frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} + \alpha_E + \mu_E + \frac{\sigma_2^2}{2}.$$

Finally, let $U_3 = \frac{1}{\theta+1}(S+E+I+V)^{\theta+1}$, where θ is a constant and satisfies:

$$0 < \theta < \frac{\mu_S \wedge \mu_E \wedge (\mu_I + \gamma_I) \wedge \mu_V}{4\theta^2(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2)}.$$

According to Equation (4), we can obtain:

$$\begin{aligned} \mathcal{L}U_3 &= \left(\Lambda - (1-\eta)(1-\mathcal{D})^\kappa \beta SI - \mu_S S - \epsilon S \right) S^\theta + \left((1-\eta)(1-\mathcal{D})^\kappa \beta SI - \alpha_E E - \mu_E E \right) E^\theta \\ &\quad + (\alpha_E E - \mu_I I - \gamma_I I) I^\theta + (\epsilon S - \mu_V V) V^\theta + \frac{1}{2} \left(\sigma_1^2 \theta S^{\theta+1} + \sigma_2^2 \theta E^{\theta+1} + \sigma_3^2 \theta I^{\theta+1} + \sigma_4^2 \theta V^{\theta+1} \right) \\ &\leq \Lambda(S+E+I+V)^\theta - \mu_S S^{\theta+1} - \mu_E E^{\theta+1} - (\mu_I + \gamma_I) I^{\theta+1} - \mu_V V^{\theta+1} \\ &\quad + \left(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2 \right) \frac{\theta}{2} (S+E+I+V)^{\theta+1}. \end{aligned}$$

Applying the inequality $|\sum_{i=1}^k c_i|^n \leq k^{n-1} \sum_{i=1}^k |c_i|^n$, $n \geq 1$, we can calculate that:

$$\begin{aligned} \mathcal{L}U_3 &\leq \Lambda(S+E+I+V)^\theta - \mu_S S^{\theta+1} - \mu_E E^{\theta+1} - (\mu_I + \gamma_I) I^{\theta+1} - \mu_V V^{\theta+1} \\ &\quad + \left(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2 \right) \frac{4^\theta \theta}{2} \left(S^{\theta+1} + E^{\theta+1} + I^{\theta+1} + V^{\theta+1} \right) \\ &= - \left(\frac{\mu_S}{4} - \frac{4^\theta \theta}{2} \left(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2 \right) \right) S^{\theta+1} - \left(\frac{\mu_E}{4} - \frac{4^\theta \theta}{2} \left(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2 \right) \right) E^{\theta+1} \\ &\quad - \left(\frac{\mu_I + \gamma_I}{4} - \frac{4^\theta \theta}{2} \left(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2 \right) \right) I^{\theta+1} - \left(\frac{\mu_V}{4} - \frac{4^\theta \theta}{2} \left(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2 \right) \right) V^{\theta+1} \\ &\quad - \frac{\mu_S}{2} S^{\theta+1} - \frac{\mu_E}{2} E^{\theta+1} - \frac{(\mu_I + \gamma_I)}{2} I^{\theta+1} - \frac{\mu_V}{2} V^{\theta+1} + \Lambda(S+E+I+V)^\theta \\ &\leq -\frac{\mu_S}{4} S^{\theta+1} - \frac{\mu_E}{4} E^{\theta+1} - \frac{(\mu_I + \gamma_I)}{4} I^{\theta+1} - \frac{\mu_V}{4} V^{\theta+1} + \Lambda(S+E+I+V)^\theta \\ &\quad - \frac{\mu_S}{2} S^{\theta+1} - \frac{\mu_E}{2} E^{\theta+1} - \frac{(\mu_I + \gamma_I)}{2} I^{\theta+1} - \frac{\mu_V}{2} V^{\theta+1} \\ &\leq -\frac{\mu_S}{2} S^{\theta+1} - \frac{\mu_E}{2} E^{\theta+1} - \frac{(\mu_I + \gamma_I)}{2} I^{\theta+1} - \frac{\mu_V}{2} V^{\theta+1} + B, \end{aligned}$$

where

$$B := \sup_{(S,E,I,V) \in \mathbb{R}_+^4} \left\{ -\frac{\mu_S}{4} S^{\theta+1} - \frac{\mu_E}{4} E^{\theta+1} - \frac{(\mu_I + \gamma_I)}{4} I^{\theta+1} - \frac{\mu_V}{4} V^{\theta+1} + \Lambda(S+E+I+V)^\theta \right\} < \infty.$$

Define a C^2 -function $U : \mathbb{R}_+^4 \rightarrow \mathbb{R}$ as follows

$$U = MU_1 + U_2 + U_3,$$

where M is a positive constant such that $-M\lambda + C \leq -2$, and

$$\begin{aligned}
C := & \sup_{(S,E,I,V) \in \mathbb{R}_+^4} \left\{ \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} \right. \\
& \left. - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - MS + M_1 \right\} \\
& \vee \sup_{(S,E,I,V) \in \mathbb{R}_+^4} \left\{ \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} \right. \\
& \left. - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - MI + M_1 \right\} \\
& \vee \sup_{(S,E,I,V) \in \mathbb{R}_+^4} \left\{ \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} \right. \\
& \left. - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - MV + M_1 \right\},
\end{aligned}$$

where $M_1 := \alpha_E + \mu_E + \frac{\sigma_2^2}{2} + B$. It is straightforward to verify that

$$\lim_{n \rightarrow \infty} \inf_{(S,E,I,V) \in \mathbb{R}_{++}^4 \setminus \mathcal{N}_k} U = +\infty.$$

Moreover, the function $U(S, E, I, V)$ within the interior of $\mathbb{R}_+^4$ possesses a minimum at the point $(\tilde{S}, \tilde{E}, \tilde{I}, \tilde{V})$. For the C^2 -function, define

$$\widehat{U} = U - U(\tilde{S}, \tilde{E}, \tilde{I}, \tilde{V}).$$

Based on Equation (4), we can obtain:

$$\begin{aligned}
\mathcal{L}\widehat{U} \leq & -M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{2} \\
& - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} + M_1.
\end{aligned}$$

Recall from Equation (5), for any $0 < \zeta < 1$,

$$\mathcal{N}_{1/\zeta} = \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : \zeta \leq S \leq \frac{1}{\zeta}, \zeta^3 \leq E \leq \frac{1}{\zeta^3}, \zeta \leq I \leq \frac{1}{\zeta}, \zeta \leq V \leq \frac{1}{\zeta} \right\},$$

we can select ζ sufficiently small to ensure the following conditions hold:

$$-\frac{\mu_S}{4\zeta^{\theta+1}} + G_1 \leq -1, \quad (6)$$

$$-\frac{\mu_E}{4\zeta^{3(\theta+1)}} + G_2 \leq -1, \quad (7)$$

$$-\frac{\mu_I + \gamma_I}{4\zeta^{\theta+1}} + G_3 \leq -1, \quad (8)$$

$$-\frac{\mu_V}{4\zeta^{\theta+1}} + G_4 \leq -1, \quad (9)$$

$$-M\lambda + M\zeta + C \leq -1, \quad (10)$$

$$-\frac{(1-\eta)(1-\mathcal{D})^\kappa \beta}{\zeta} + G_5 \leq -1, \quad (11)$$

where G_1, G_2, G_3, G_4, G_5 are all positive constants, explicitly defined in Equations (6)–(11), respectively. We partition $\mathcal{N}_{1/\zeta}^\zeta := \mathbb{R}_+^4 \setminus \mathcal{N}_{1/\zeta}$ into the following eight domains to demonstrate that $\mathcal{L}\widehat{U} \leq -1$ on $\mathcal{N}_{1/\zeta}$:

$$\begin{aligned}
\mathcal{H}_1 &= \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : S > \frac{1}{\zeta} \right\}, \mathcal{H}_2 = \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : E > \frac{1}{\zeta^3} \right\}, \\
\mathcal{H}_3 &= \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : I > \frac{1}{\zeta} \right\}, \mathcal{H}_4 = \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : V > \frac{1}{\zeta} \right\}, \\
\mathcal{H}_5 &= \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : 0 < I < \zeta \right\}, \mathcal{H}_6 = \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : 0 < V < \zeta \right\}, \\
\mathcal{H}_7 &= \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : 0 < S < \zeta \right\}, \mathcal{H}_8 = \left\{ (S, E, I, V) \in \mathbb{R}_+^4 : \zeta < S, 0 < E < \zeta^3, \zeta < I \right\}.
\end{aligned}$$

Case1. For every $(S, E, I, V) \in \mathcal{H}_1$, we have:

$$\begin{aligned}
\mathcal{S}\tilde{U} &\leq -\frac{\mu_S S^{\theta+1}}{4} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I \\
&\quad - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{4} + M_1 \\
&\leq -\frac{\mu_S}{4\zeta^{\theta+1}} + G_1,
\end{aligned}$$

where.

$$G_1 := \sup_{(S, E, I, V) \in \mathbb{R}_+^4} \left\{ -\frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I \right. \\
\left. - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{4} + M_1 \right\}.$$

By virtue of Equation (6), it is confirmed that $\mathcal{S}\tilde{U} \leq -1$.

Case2. For every $(S, E, I, V) \in \mathcal{H}_2$, we have:

$$\begin{aligned}
\mathcal{S}\tilde{U} &\leq -\frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{4} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I \\
&\quad - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_E E^{\theta+1}}{4} + M_1 \\
&\leq -\frac{\mu_E}{4\zeta^{3(\theta+1)}} + G_2,
\end{aligned}$$

where

$$G_2 := \sup_{(S, E, I, V) \in \mathbb{R}_+^4} \left\{ -\frac{\mu_S S^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I \right. \\
\left. - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_E E^{\theta+1}}{4} + M_1 \right\}.$$

By virtue of Equation (7), it is confirmed that $\mathcal{S}\tilde{U} \leq -1$.

Case3. For every $(S, E, I, V) \in \mathcal{H}_3$, we have:

$$\begin{aligned}
\mathcal{S}\tilde{U} &\leq -\frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{4} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I \\
&\quad - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_I + \gamma_I}{4} I^{\theta+1} + M_1 \\
&\leq -\frac{\mu_I + \gamma_I}{4\zeta^{\theta+1}} + G_3,
\end{aligned}$$

where

$$G_3 := \sup_{(S,E,I,V) \in \mathbb{R}_+^4} \left\{ -\frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{4} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I \right. \\ \left. - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{2} + M_1 \right\}.$$

By virtue of Equation (8), it is confirmed that $\mathcal{L}\tilde{U} \leq -1$.

Case4. For every $(S, E, I, V) \in \mathcal{H}_4$, we have:

$$\mathcal{L}\tilde{U} \leq -\frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{4} - \frac{\mu_V V^{\theta+1}}{4} - M\lambda + \frac{MV}{S^2} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I \\ - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_V V^{\theta+1}}{4} + M_1 \\ \leq -\frac{\mu_V}{4\zeta^{\theta+1}} + G_4,$$

where

$$G_4 := \sup_{(S,E,I,V) \in \mathbb{R}_+^4} \left\{ -\frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{4} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - M\lambda + \frac{MV}{S^2} \right. \\ \left. - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} - \frac{\mu_S S^{\theta+1}}{2} + M_1 \right\}.$$

By virtue of Equation (9), it is confirmed that $\mathcal{L}\tilde{U} \leq -1$.

Case5. For every $(S, E, I, V) \in \mathcal{H}_5$, we have:

$$\mathcal{L}\tilde{U} \leq -\frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{4} - M\lambda + \frac{MV}{S^2} - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} \\ - \frac{\mu_V V^{\theta+1}}{4} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - MI + MI + M_1 \\ \leq -M\lambda + M\zeta + C.$$

According to (10) and $-M\lambda + C \leq -2$, it is confirmed that $\mathcal{L}\tilde{U} \leq -1$.

Case6. For every $(S, E, I, V) \in \mathcal{H}_6$, we have:

$$\mathcal{L}\tilde{U} \leq -\frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{4} - M\lambda + \frac{MV}{S^2} - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} \\ - \frac{\mu_V V^{\theta+1}}{4} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - MV + MV + M_1 \\ \leq -M\lambda + M\zeta + C.$$

According to (10) and $-M\lambda + C \leq -2$, it is confirmed that $\mathcal{L}\tilde{U} \leq -1$.

Case7. For every $(S, E, I, V) \in \mathcal{H}_7$, we have:

$$\mathcal{L}\tilde{U} \leq -\frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I) I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{4} - M\lambda + \frac{MV}{S^2} - \frac{(1-\eta)(1-\mathcal{D})^\kappa \beta SI}{E} \\ - \frac{\mu_V V^{\theta+1}}{4} + M(1-\eta)(1-\mathcal{D})^\kappa \beta I - MS + MS + M_1 \\ \leq -M\lambda + M\zeta + C.$$

According to (10) and $-M\lambda + C \leq -2$, it is confirmed that $\mathcal{L}\tilde{U} \leq -1$.

Case8. For every $(S, E, I, V) \in \mathcal{H}_8$, we have:

$$\begin{aligned}\mathcal{L}\tilde{U} &\leq -\frac{(1-\eta)(1-\mathcal{D})^\kappa\beta SI}{E} - \frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I)I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M_1 \\ &\leq -\frac{(1-\eta)(1-\mathcal{D})^\kappa\beta}{\zeta} + G_5,\end{aligned}$$

where

$$G_5 := \sup_{(S,E,I,V) \in \mathbb{R}_+^4} \left\{ -\frac{\mu_S S^{\theta+1}}{2} - \frac{\mu_E E^{\theta+1}}{2} - \frac{(\mu_I + \gamma_I)I^{\theta+1}}{2} - \frac{\mu_V V^{\theta+1}}{2} - M\lambda + \frac{MV}{S^2} + M_1 \right\}.$$

By virtue of Equation (11), it is confirmed that $\mathcal{L}\tilde{U} \leq -1$.

We have demonstrated that for sufficiently small $\zeta > 0$, $\mathcal{L}\tilde{U} \leq -1$ holds for each $(S, E, I, V) \in \mathbb{R}_+^4 \setminus \mathcal{N}_{1/\zeta}$ and Theorem 2 is valid.

2.3. Extinction

This subsection is dedicated to determining the sufficient conditions for disease extinction in the model (3).

Theorem 3. Let Q_t be the solution to the model (3) with starting point $Q_0 \in \mathbb{R}_+^4$, and denote

$$M := \frac{\mu_I + \gamma_I}{C} \mathbb{J}_2^{\frac{1}{2}} + \mathbb{J}_1^{\frac{1}{2}} - \frac{1}{2} \xi \left(\sigma_2^{-2} + \sigma_3^{-2} + \sigma_4^{-2} + 1 \right)^{-1},$$

where $\mathbb{J}_1$, $\mathbb{J}_2$, and C are constants, ξ is a constant between 0 and 1, and it satisfies

$$\xi := \inf_{(E,I,V) \in \mathbb{R}_+^3} \left\{ \frac{\sigma_2^2 E^2 + \theta_1^2 \sigma_3^2 I^2 + \sigma_4^2 V^2}{\sigma_2^2 E^2 + \theta_1^2 \sigma_3^2 I^2 + \sigma_4^2 V^2 + \mathbb{A}^2} \right\}.$$

Assume $m > M$, where m is specified in Equation (18). Then

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \ln(E(t) + \theta_1 I(t) + V(t)) < 0 \text{ a.s.,}$$

where θ_1 is a positive constant given by Equation (17). Proof From the S compartment equation in model (3), we have

$$\begin{aligned}dS(t) &= \left\{ \Lambda - (1-\eta)(1-\mathcal{D})^\kappa \beta S(t)I(t) - \mu_S S(t) - \epsilon S(t) \right\} dt + \sigma_1 S(t) dB_1 \\ &\leq \left\{ \Lambda - (\mu_S + \epsilon) S(t) \right\} dt + \sigma_1 S(t) dB_1.\end{aligned}$$

Consider the following equation with a stochastic disturbance

$$dX(t) = \{ \Lambda - (\mu_S + \epsilon) X(t) \} dt + \sigma_1 X(t) dB_1. \quad (12)$$

Let $h(x) = \Lambda - (\mu_S + \epsilon)x$, $\sigma(x) = x\sigma_1$, we can compute that:

$$\int_0^\infty \frac{h(x)}{\sigma^2(x)} dx = \frac{1}{\sigma_1^2} \int_0^\infty \left(\frac{\Lambda}{x^2} - \frac{\mu_S + \epsilon}{x} \right) dx = \frac{1}{\sigma_1^2} \left\{ -\frac{\Lambda}{x} - (\mu_S + \epsilon) \ln x \right\} + J.$$

Thus,

$$e^{\int_0^\infty \frac{h(x)}{\sigma^2(x)} dx} = e^J x^{-\frac{\mu_S + \epsilon}{\sigma_1^2}} e^{-\frac{\Lambda}{\sigma_1^2 x}},$$

we can obtain

$$\int_0^\infty \frac{1}{\sigma^2(x)} e^{\int_1^x \frac{2h(\tau)}{\sigma^2(\tau)} d\tau} dx = C \int_0^\infty x^{-2} x^{-\frac{2(\mu_S + \epsilon)}{\sigma_1^2}} e^{-\frac{2\Lambda}{\sigma_1^2 x}} dx = C \int_0^\infty x^{-2} x^{-\frac{2(\mu_S + \epsilon)}{\sigma_1^2}} e^{-\frac{2\Lambda}{\sigma_1^2 x}} dx \leq \infty, \quad (13)$$

where $C = \frac{e^{2J}}{\sigma_1^2}$ is a constant, and

$$\begin{aligned}
\frac{1}{\sigma_1^2} \int_0^\infty x^{-2-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} e^{-\frac{2\Lambda}{\sigma_1^2}x} dx &= \frac{1}{\sigma_1^2} \int_0^\infty x^{-2-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} e^{-\frac{2\Lambda}{\sigma_1^2}x} dx \\
&= \frac{1}{\sigma_1^2} \int_0^\infty \left(\frac{2\Lambda}{\sigma_1^2}y\right)^{-2-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} e^{-y} \left(\frac{2\Lambda}{\sigma_1^2}y^2\right) dy \\
&= \frac{1}{\sigma_1^2} \left(\frac{2\Lambda}{\sigma_1^2}\right)^{-2-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} \frac{2\Lambda}{\sigma_1^2} \int_0^\infty y^{\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} e^{-y} dy \\
&= \frac{1}{\sigma_1^2} \left(\frac{2\Lambda}{\sigma_1^2}\right)^{-1-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} \int_0^\infty y^{\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} e^{-y} dy \\
&= \frac{1}{\sigma_1^2} \left(\frac{2\Lambda}{\sigma_1^2}\right)^{-1-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} \Gamma\left(1 + \frac{2(\mu_s+\epsilon)}{\sigma_1^2}\right).
\end{aligned}$$

Here, Γ denotes the Gamma function. Consequently, Equation (13) meets the criteria outlined in (Kutoyants, 2004), implying that the system in (12) is ergodic with a sole stationary distribution characterized by the probability density function

$$\pi(x) = \mathbb{J} \sigma_1^{-2} x^{-2-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} e^{-\frac{2\Lambda}{\sigma_1^2}x}, \quad x \in (0, \infty),$$

where

$$\mathbb{J} = \left(\frac{1}{\sigma_1^2} \left(\frac{2\Lambda}{\sigma_1^2}\right)^{-1-\frac{2(\mu_s+\epsilon)}{\sigma_1^2}} \Gamma\left(1 + \frac{2(\mu_s+\epsilon)}{\sigma_1^2}\right) \right)^{-1}$$

is a constant and ensures that $\int_0^\infty \pi(x) dx = 1$. According to the ergodicity, it follows that:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t X(\tau) d\tau = \int_0^\infty x \pi(x) dx \quad \text{a.s.} \quad (14)$$

When $X(0) = S(0) > 0$, the comparison theorem for stochastic differential equations yields

$$S(t) \leq X(t) \quad \text{for all } t \geq 0 \quad \text{a.s.} \quad (15)$$

Define a twice-differentiable mapping $\mathbb{V}_1 : \mathbb{R}_{++}^3 \rightarrow \mathbb{R}_{++}$ as follows:

$$\mathbb{V}_1 = E + \theta_1 I + V + \mathbb{A},$$

where θ_1 denotes a positive parameter and $\mathbb{A} \geq \epsilon$. Set $\mathbb{V}_1(t) = E(t) + \theta_1 I(t) + V(t) + \mathbb{A}$. Applying Itô's lemma, with S representing $S(t)$ and analogous notation for other variables, yields

$$d(\ln \mathbb{V}_1(t)) = \mathcal{L}(\ln \mathbb{V}_1(t)) dt + \frac{1}{\mathbb{V}_1(t)} (\sigma_2 E dB_2(t) + \sigma_3 \theta_1 I dB_3(t) + \sigma_4 V dB_4(t)),$$

in which

$$\begin{aligned}
\mathcal{L}(\ln \mathbb{V}_1(t)) &= \frac{1}{\mathbb{V}_1(t)} ((1-\eta)(1-\mathcal{D})^\mathcal{K} \beta SI - \alpha_E E - \mu_E E + \theta_1 (\alpha_E E - \mu_I I - \gamma_I I) \\
&\quad + (\epsilon S - \mu_V V) - \frac{\sigma_2^2 E^2}{2\mathbb{V}_1(t)^2} - \frac{\theta_1 \sigma_3^2 I^2}{2\mathbb{V}_1(t)^2} - \frac{\sigma_4^2 V^2}{2\mathbb{V}_1(t)^2}.
\end{aligned}$$

Moreover,

$$\begin{aligned}\mathbb{V}_1(t)^2 &= \left(\sigma_2 E \frac{1}{\sigma_2} + \theta_1 \sigma_3 I \frac{1}{\sigma_3} + \sigma_4 V \frac{1}{\sigma_4} + \mathbb{A} \right)^2 \\ &\leq \left(\sigma_2^2 E^2 + \theta_1^2 \sigma_3^2 I^2 + \sigma_4^2 V^2 + \mathbb{A}^2 \right) \left(\frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2} + \frac{1}{\sigma_4^2} + 1 \right).\end{aligned}$$

And

$$\begin{aligned}& \frac{1}{\mathbb{V}_1(t)} \left((1-\eta)(1-\mathcal{D})^\kappa \beta SI - \alpha_E E - \mu_E E + \theta_1 (\alpha_E E - \mu_I I - \gamma_I I) + (\epsilon S - \mu_V V) \right) \\ & \leq \frac{1}{\mathbb{V}_1(t)} (1-\eta)(1-\mathcal{D})^\kappa \beta I (S - \mathbb{C}) + \frac{1}{\mathbb{V}_1(t)} \left((1-\eta)(1-\mathcal{D})^\kappa \mathbb{A} \beta I - \theta_1 \mu_I I - \theta_1 \gamma_I I \right) \\ & \quad + \frac{1}{\mathbb{V}_1(t)} (\theta_1 \alpha_E E - \alpha_E E - \mu_E E) + \frac{1}{\mathbb{V}_1(t)} \epsilon S \\ & \leq \frac{1}{\theta_1} (1-\eta)(1-\mathcal{D})^\kappa \beta |X - \mathbb{C}| + |X| + \frac{I}{\mathbb{V}_1(t)} \left((1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta - \theta_1 (\mu_I + \gamma_I) \right) \\ & \quad + \frac{E}{\mathbb{V}_1(t)} (\theta_1 \alpha_E - (\alpha_E + \mu_E)),\end{aligned}\tag{16}$$

where $\mathbb{C}$ satisfies

$$\frac{(1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta}{\mu_I + \gamma_I} \leq \theta_1 \leq \frac{\alpha_E + \mu_E}{\alpha_E}.\tag{17}$$

Next, we need a suitable constant m to satisfy

$$(1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta - \theta_1 (\mu_I + \gamma_I) \leq -\theta_1 m, \quad \theta_1 \alpha_E - (\alpha_E + \mu_E) \leq -m.$$

Then, we can get

$$m \leq \mu_I + \gamma_I - \frac{\alpha_E (1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta}{\alpha_E + \mu_E}, \quad m \leq \alpha_E + \mu_E - \frac{\alpha_E (1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta}{\mu_I + \gamma_I}.$$

Therefore, we choose

$$m = \min \left\{ \mu_I + \gamma_I - \frac{\alpha_E (1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta}{\alpha_E + \mu_E}, \alpha_E + \mu_E - \frac{\alpha_E (1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta}{\mu_I + \gamma_I} \right\}.\tag{18}$$

Then Equation (16) can be simplified as

$$\begin{aligned}& \frac{1}{\mathbb{V}_1(t)} \left((1-\eta)(1-\mathcal{D})^\kappa \beta SI - \alpha_E E - \mu_E E + \theta_1 (\alpha_E E - \mu_I I - \gamma_I I) + (\epsilon S - \mu_V V) \right) \\ & \leq \frac{1}{\theta_1} (1-\eta)(1-\mathcal{D})^\kappa \beta |X - \mathbb{C}| + |X| - m.\end{aligned}$$

Then,

$$\begin{aligned}\mathcal{L}(\ln \mathbb{V}_1(t)) &\leq \frac{1}{\theta_1} (1-\eta)(1-\mathcal{D})^\kappa \beta |X - \mathbb{C}| + |X| - m - \frac{\sigma_2^2 E^2}{2 \mathbb{V}_1(t)^2} - \frac{\theta_1 \sigma_3^2 I^2}{2 \mathbb{V}_1(t)^2} - \frac{\sigma_4^2 V^2}{2 \mathbb{V}_1(t)^2} \\ &\leq \frac{1}{\theta_1} (1-\eta)(1-\mathcal{D})^\kappa \beta |X - \mathbb{C}| + |X| - m - \frac{1}{2} \xi \left(\frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2} + \frac{1}{\sigma_4^2} + 1 \right)^{-1}.\end{aligned}\tag{19}$$

Following integration of Equation (19) over the interval from 0 to t and subsequent division of both sides by t , the following inequality holds:

$$\begin{aligned}
\frac{\ln \mathbb{V}_1(t)}{t} &\leq \frac{\ln \mathbb{V}_1(0)}{t} - m + \frac{\beta(1-\eta)(1-\mathcal{D})^\kappa}{\theta_1 t} \int_0^t |X(\tau) - \mathbb{C}| d\tau - \frac{1}{2} \xi \left(\frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2} + \frac{1}{\sigma_4^2} + 1 \right)^{-1} \\
&\quad + \frac{1}{t} \int_0^t \frac{\sigma_2 E(\tau)}{\mathbb{V}_1(\tau)} dB_2(\tau) + \frac{1}{t} \int_0^t \frac{\theta_1 \sigma_3 I(\tau)}{\mathbb{V}_1(\tau)} dB_3(\tau) + \frac{1}{t} \int_0^t \frac{\sigma_4 V(\tau)}{\mathbb{V}_1(\tau)} dB_4(\tau) + \frac{1}{t} \int_0^t |X(\tau)| d\tau \\
&= \frac{\ln \mathbb{V}_1(0)}{t} - m + \frac{\beta(1-\eta)(1-\mathcal{D})^\kappa}{\theta_1 t} \int_0^t |X(\tau) - \mathbb{C}| d\tau - \frac{1}{2} \xi \left(\frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2} + \frac{1}{\sigma_4^2} + 1 \right)^{-1} \\
&\quad + \frac{M_2(t)}{t} + \frac{M_3(t)}{t} + \frac{M_4(t)}{t} + \frac{1}{t} \int_0^t |X(\tau)| d\tau.
\end{aligned}$$

We define the processes as follows:

$$M_2(t) = \int_0^t \frac{\sigma_2 E(\tau)}{\mathbb{V}_1(\tau)} dB_2(\tau), \quad M_3(t) = \int_0^t \frac{\theta_1 \sigma_3 I(\tau)}{\mathbb{V}_1(\tau)} dB_3(\tau), \quad M_4(t) = \int_0^t \frac{\sigma_4 V(\tau)}{\mathbb{V}_1(\tau)} dB_4(\tau).$$

We observe that the sequences $\{M_2(t)\}_{t \geq 0}$, $\{M_3(t)\}_{t \geq 0}$, and $\{M_4(t)\}_{t \geq 0}$ are local martingales with quadratic variations satisfying the following bounds:

$$\begin{aligned}
\langle M_2, M_2 \rangle_t &= \sigma_2^2 \int_0^t \left(\frac{E(\tau)}{\mathbb{V}_1(\tau)} \right)^2 d\tau \leq \sigma_2^2 t, \quad \langle M_3, M_3 \rangle_t = \sigma_3^2 \int_0^t \left(\frac{\theta_1 I(\tau)}{\mathbb{V}_1(\tau)} \right)^2 d\tau \leq \sigma_3^2 t, \\
\langle M_4, M_4 \rangle_t &= \sigma_4^2 \int_0^t \left(\frac{V(\tau)}{\mathbb{V}_1(\tau)} \right)^2 d\tau \leq \sigma_4^2 t.
\end{aligned}$$

Applying the strong law of large numbers as stated in (Mao, 2007), it follows that $\lim_{t \rightarrow \infty} M_i(t)/t = 0$, $i = 2, 3, 4$, a.s. Given the ergodic property of the process $\{X(t)\}_{t \geq 0}$ and the finiteness of $\int_0^\infty |x - \mathbb{C}| \pi(x) dx < \infty$, we deduce that

$$\begin{aligned}
\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t |X(\tau) - \mathbb{C}| d\tau &= \int_{-\infty}^\infty |x - \mathbb{C}| \pi(x) dx \leq \left(\int_{-\infty}^\infty (x - \mathbb{C})^2 \pi(x) dx \right)^{1/2}, \\
\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t |X(\tau)| d\tau &= \int_{-\infty}^\infty |x| \pi(x) dx \leq \left(\int_{-\infty}^\infty x^2 \pi(x) dx \right)^{1/2},
\end{aligned}$$

where

$$\begin{aligned}
\int_0^t x \pi(x) dx &= \frac{2\Lambda}{\sigma_1^2} \left(1 + \frac{2(\mu_s + \epsilon)}{\sigma_1^2} \right), \\
\int_0^t x^2 \pi(x) dx &= \left(\frac{2\Lambda}{\sigma_1^2} \right)^2 \left(1 + \frac{2(\mu_s + \epsilon)}{\sigma_1^2} \right) \left(2 + \frac{2(\mu_s + \epsilon)}{\sigma_1^2} \right) = \mathbb{J}_1, \\
\int_0^t (x - \mathbb{C})^2 \pi(x) dx &= \frac{2\Lambda}{\sigma_1^2} \left(1 + \frac{2(\mu_s + \epsilon)}{\sigma_1^2} \right) \left(\frac{4\Lambda}{\sigma_1^2} + \frac{4\Lambda(\mu_s + \epsilon)}{\sigma_1^4} - 2\mathbb{C} \right) + \mathbb{C}^2 = \mathbb{J}_2.
\end{aligned}$$

Then we can get

$$\limsup_{t \rightarrow \infty} \frac{\ln \mathbb{V}_1(t)}{t} \leq -m + \frac{\beta(1-\eta)(1-\mathcal{D})^\kappa}{\theta_1} \mathbb{J}_2 + \mathbb{J}_1^2 - \frac{1}{2} \xi \left(\sigma_2^{-2} + \sigma_3^{-2} + \sigma_4^{-2} + 1 \right)^{-1}.$$

Since $\frac{(1-\eta)(1-\mathcal{D})^\kappa \mathbb{C} \beta}{\mu_l + \gamma_l} \leq \theta_1 \leq \frac{\alpha_E + \mu_E}{\alpha_E}$, we can get

$$\limsup_{t \rightarrow \infty} \frac{\ln \mathbb{V}_1(t)}{t} \leq -m + \frac{\mu_l + \gamma_l}{\mathbb{C}} \mathbb{J}_2 + \mathbb{J}_1^2 - \frac{1}{2} \xi \left(\sigma_2^{-2} + \sigma_3^{-2} + \sigma_4^{-2} + 1 \right)^{-1}.$$

With the constant $M = \frac{\mu_l + \gamma_l}{\mathbb{C}} \mathbb{J}_2 + \mathbb{J}_1^2 - \frac{1}{2} \xi \left(\sigma_2^{-2} + \sigma_3^{-2} + \sigma_4^{-2} + 1 \right)^{-1}$, it follows that

$$\limsup_{t \rightarrow \infty} \frac{\ln \mathbb{V}_t(t)}{t} \leq -m + M < 0. \quad (20)$$

Then, we can draw the conclusion. $\limsup_{t \rightarrow \infty} \frac{1}{t} \ln(E(t) + \theta_1 I(t) + V(t)) < 0$ a.s.

3. Numerical simulation

We represent S_n , E_n , I_n , and V_n as the respective counts of susceptible, exposed, infected, and vaccinated individuals at time step t . To derive an approximate solution over the temporal domain $[0, t]$, the interval is partitioned into l uniform subintervals with a step size of $\Delta t = t/l$, and the increment $\Delta B_n = B(t_{n+1}) - B(t_n)$ is defined, with $t_n = n\Delta t$. The discretization is performed employing the Milstein higher-order Method (Milstein, 1974), and the model (3) be

$$\begin{cases} S_{n+1} = S_n + [\Lambda - (1-\eta)(1-D)^{\mathcal{K}}\beta S_n I_n - \mu_S S_n - \epsilon S_n] \Delta t + \sigma_1 S_n \Delta B_{1,n} + \frac{1}{2} \sigma_1^2 S_n ((\Delta B_{1,n})^2 - \Delta t), \\ E_{n+1} = E_n + [(1-\eta)(1-D)^{\mathcal{K}}\beta S_n I_n - \alpha_E E_n - \mu_E E_n] \Delta t + \sigma_2 E_n \Delta B_{2,n} + \frac{1}{2} \sigma_2^2 E_n ((\Delta B_{2,n})^2 - \Delta t), \\ I_{n+1} = I_n + [\alpha_E E_n - \mu_I I_n - \gamma I_n] \Delta t + \sigma_3 I_n \Delta B_{3,n} + \frac{1}{2} \sigma_3^2 I_n ((\Delta B_{3,n})^2 - \Delta t), \\ V_{n+1} = V_n + [\epsilon S_n - \mu_V V_n] \Delta t + \sigma_4 V_n \Delta B_{4,n} + \frac{1}{2} \sigma_4^2 V_n ((\Delta B_{4,n})^2 - \Delta t). \end{cases} \quad (21)$$

Example 1. Assume that a disease outbreak occurs at the initial time $t = 0$, where the S compartment constitutes 94% of the total population, the E compartment accounts for 5%, and the I compartment represents 1%. The model parameters are set as $\Lambda = 0.05$, $\eta = 0.5$, $\mathcal{D} = 0.5$, $\mathcal{K} = 1$, $\beta = 0.5$, $\mu_S = 0.001$, $\mu_E = 0.005$, $\mu_I = 0.005$, $\mu_V = 0.001$, $\epsilon = 0.001$, $\alpha = 0.03$, $\gamma = 0.01$, $\sigma_1 = 0.03$, $\sigma_2 = 0.03$, $\sigma_3 = 0.03$, $\sigma_4 = 0.03$, with $R_0^S > 1$ at this stage, which satisfies the condition in Theorem 2.

Fig. 2 represents the time series plots of the solutions $S(t)$, $E(t)$, $I(t)$ and $V(t)$ for model (3). It is evident that, as time progresses, the values gradually stabilize within certain ranges and do not converge to zero, indicating that the model is ergodic and suggesting the persistent existence of the disease.

We establish the parameter values $\eta = 0.6$, $\mathcal{D} = 0.6$, and $\mathcal{K} = 1$, taking into account the fundamental pandemic response capacity of the social healthcare system as well as the public awareness and preventive measures against infectious diseases. These parameters encapsulate the role of medical resources and public health interventions in managing epidemics, where η reflects the reduction in contact rates due to governmental interventions (such as isolation measures), $\mathcal{D}$ denotes the dynamic control of disease transmission through public self-protective behaviors (e.g., mask-wearing and social distancing), and $\mathcal{K}$ represents the typical response level of the public. This parameterization assumes that society possesses a certain level of healthcare infrastructure and educational outreach, effectively reducing the risk of transmission.

Example 2. Take $\Lambda = 0.05$, $\eta = 0.9$, $\mathcal{D} = 0.9$, $\mathcal{K} = 2$, $\beta = 0.5$, $\mu_S = 0.001$, $\mu_E = 0.005$, $\mu_I = 0.005$, $\mu_V = 0.001$, $\epsilon = 0.001$, $\alpha = 0.03$, $\gamma = 0.01$, $\sigma_1 = 0.03$, $\sigma_2 = 0.03$, $\sigma_3 = 0.03$, $\sigma_4 = 0.03$. Other conditions are similar to those of Example 1. Thus, $R_0^S \leq 1$ at this stage, which satisfies the parameter condition in Theorem 3.

Analyzing Fig. 3, we observe that $E(t)$ and $I(t)$ ultimately converge to zero, with the exception of $S(t)$ and $V(t)$. This indicates that, at this time point, the populations $E(t)$ and $I(t)$ will disappear, and the entire population will consist of the susceptible population $S(t)$ and the vaccinated population $V(t)$. Consequently, Fig. 4 demonstrates that the disease will eventually disappear, and the stochastic model exhibits extinction, thereby proving Theorem 3.

In the process of conducting numerical simulations, we employ the control variable method, systematically modifying the parameters η , $\mathcal{D}$, and $\mathcal{K}$ to investigate the impact of governmental interventions, public self-protective behaviors, and the level of societal awareness regarding infectious diseases on epidemic transmission. These parameters respectively represent the intensity of governmental intervention, the dynamic control capacity of public self-protective actions, and the degree of societal response to the disease. By isolating the effects of these target parameters while holding other variables constant, we were able to focus on their specific influences. Simulations demonstrate that integrated governmental interventions and public protective measures effectively interrupt epidemic transmission and reduce infections, providing a theoretical foundation for public health policy.

4. Stochastic physics-informed neural networks based on SEIRV model

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. (Raissi et al., 2019), embed physical laws into deep learning by computing partial differential equation residuals through automatic differentiation and incorporating them into the loss function. This enables simultaneous fitting to sparse observational data and the strict satisfaction of governing equations, achieving effective data-physics integration. PINNs provide mesh-free discretization, low data requirements, and

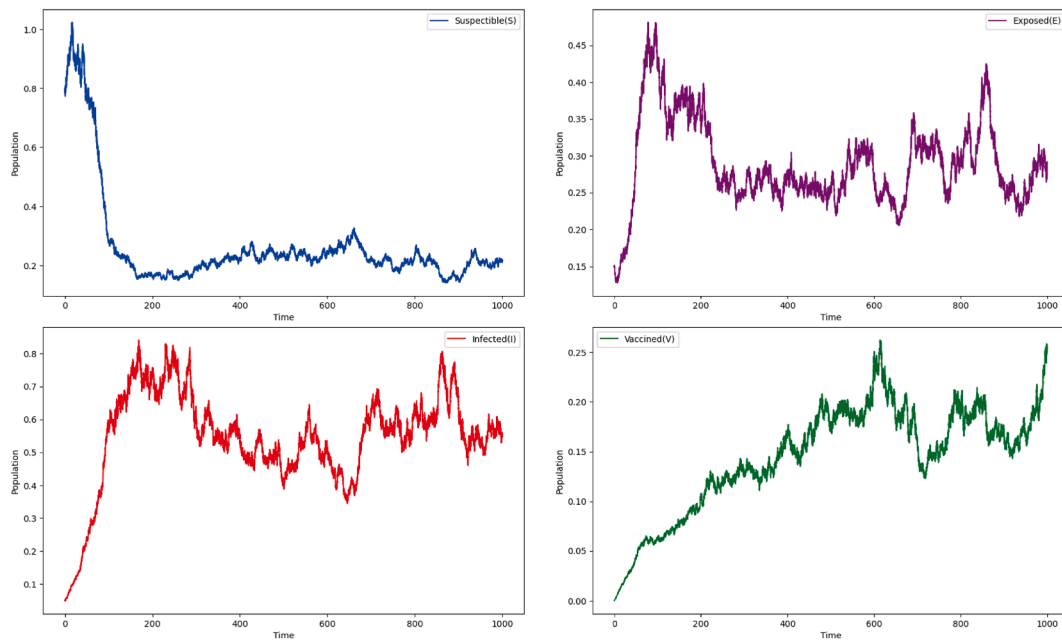


Fig. 2. Time series simulation of the stochastic SEIRV model demonstrating ergodicity, with parameters as specified in [Example 1](#).

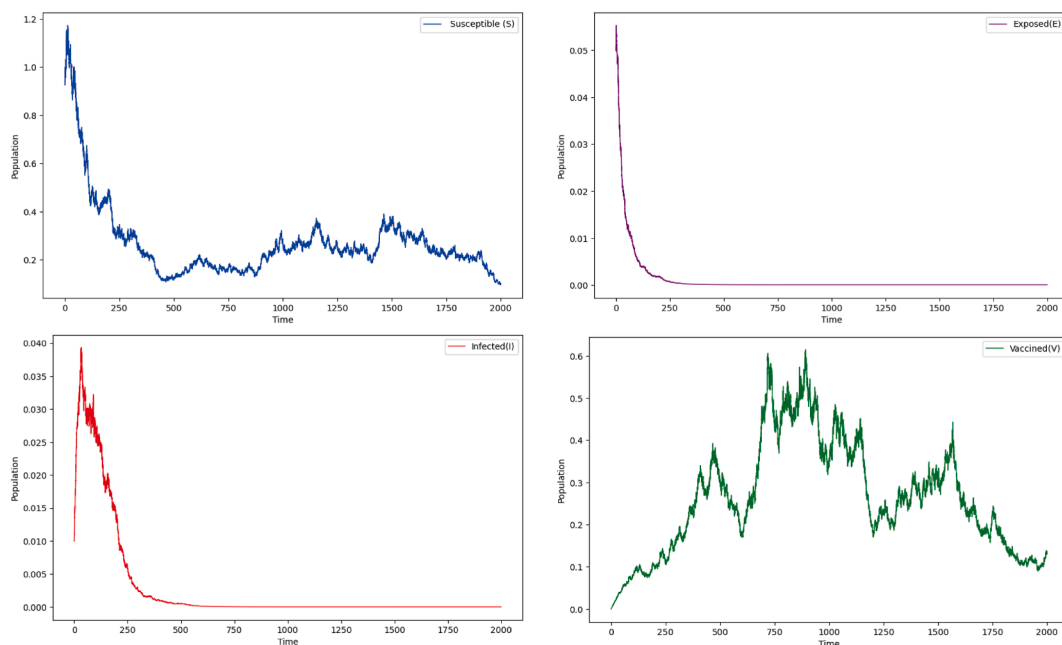


Fig. 3. Time series simulation of the stochastic SEIRV model demonstrating Extinction, with parameters as specified in [Example 2](#).

unified handling of forward and inverse problems. However, standard PINNs display clear limitations under strong noise. Mean-squared error losses are highly sensitive to non-Gaussian noise and outliers, easily causing noise overfitting. Data and physics residual terms often exhibit severe imbalance, especially in inverse problems, leading to biased parameter estimates. Automatic differentiation often amplifies noise, resulting in spurious oscillations or nonphysical solution behavior. Training stability decreases markedly, and convergence becomes difficult. Although PINNs show great promise in domains with well-defined physical priors and scarce data, substantial enhancements are still essential to ensure reliable performance in engineering applications dominated by complex, realistic noise.

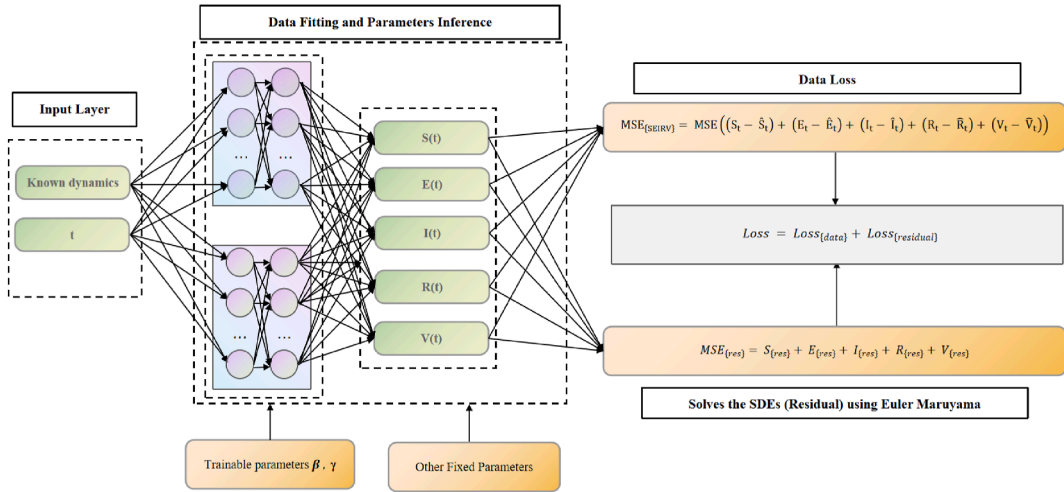


Fig. 4. Schematic diagram of S-PINNs. The black dashed section comprises a dense neural network that takes time t and the dynamic system of stochastic differential equations as inputs, predicting the populations of susceptible (S), exposed (E), infected (I), recovered (R), and vaccinated (V) individuals. Trainable parameters include the transmission rate β and recovery rate γ , while fixed parameters represent vaccination rate ε , natural mortality rates for each compartment specified as $\mu_S, \mu_E, \mu_I, \mu_R$, and μ_V , the progression rate of exposed individuals to infectious state α_E , and noise levels $\sigma_i (i = 1, 2, 3, 4, 5)$. Each compartment S, E, I, R, V is perturbed by different choices of σ values. The Equation MSE_{SEIRV} displays the residual calculation that captures deviations from the expected behavior predicted by the stochastic vaccine model. The Euler-Maruyama technique is employed to discretize the SDE. The loss function incorporates both data mismatch and physical residuals to optimize network performance. By minimizing the loss function, S-PINNs fits observed data, infers dynamic parameters β and γ , and satisfies the dynamical equations of the stochastic SEIRV model. This integration enables accurate modeling and prediction of population dynamics in epidemiology.

Stochastic Physics-Informed Neural Networks (S-PINNs) is an extension of the PINNs algorithm that incorporates noise consideration. This algorithm innovatively integrates the powerful approximation capabilities of deep learning with stochastic process modeling, effectively capturing and quantifying the inherent stochastic dynamics and uncertainties in stochastic differential equations. Next, we integrate S-PINNs with the stochastic SEIRV model.

In the initialization phase, we employ a randomization strategy to configure the neural network's weight matrices and bias vectors, as well as the epidemic model's transmission rate β and recovery rate γ . These parameters constitute the fundamental elements of the entire learning framework and are continuously refined through optimization algorithms during subsequent iterative training procedures. Let w be the neural network weights and b be the bias. The training procedure adopts a multi-layered, nested-loop architecture. The outer loop governs the total number of training epochs, where each epoch represents a comprehensive traversal and learning phase over the complete training dataset. This design ensures that the model can fully utilize all available training information. Within each training epoch, the algorithm implements a specifically designed inner loop, whose iteration count is determined by the Monte Carlo sampling number N_{MC} (Galasso et al., 2024). The primary function of this inner loop is to systematically capture and model various stochastic factors and uncertainty sources inherent in epidemic transmission processes by introducing different random noise perturbations and repeatedly executing numerical simulations of the stochastic epidemic model.

During each Monte Carlo iteration, the algorithm utilizes the current time point t as an input variable, processing it through the neural network's forward propagation mechanism to ultimately generate numerical predictions for the state variables in the model, including key epidemiological indicators such as S, E, I, R , and V . This can be mathematically formulated as:

$$S, E, I, R, V = \text{NN}(t; w, b).$$

Here, NN denotes the forward propagation function of the neural network in the black dashed section of Fig. 4. The parameters w and b are initialized using Xavier and zero initializations in TensorFlow. Subsequently, the algorithm applies Euler-Maruyama discretization to compute the subsequent values of S, E, I, R , and V based on stochastic differential equation terms and stochastic noise. The discrete update equations for each compartment are expressed as:

$$\begin{cases} S_{t+1} = S_t + \left(\Lambda - (1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \mu_S S(t) - \varepsilon S(t) \right) \Delta t + \sigma_1 \sqrt{\Delta t} S_t dB_t, \\ E_{t+1} = E_t + \left((1 - \eta)(1 - \mathcal{D})^\kappa \beta S(t)I(t) - \alpha_E E(t) - \mu_E E(t) \right) \Delta t + \sigma_2 \sqrt{\Delta t} E_t dB_t, \\ I_{t+1} = I_t + \left(\alpha_E E(t) - \mu_I I(t) - \gamma I(t) \right) \Delta t + \sigma_3 \sqrt{\Delta t} I_t dB_t, \\ R_{t+1} = R_t + \left(\gamma I(t) - \mu_R R(t) \right) \Delta t + \sigma_4 \sqrt{\Delta t} R_t dB_t, \\ V_{t+1} = V_t + \left(\varepsilon S(t) - \mu_V V(t) \right) \Delta t + \sigma_5 \sqrt{\Delta t} V_t dB_t. \end{cases}$$

The algorithm computes the discrete loss MSE_{SDE} for each temporal interval. This loss quantifies the variance among consecutive values of S, E, I, R , and V , facilitating the assessment of approximation error when discretizing continuous-time SDEs via the Euler-Maruyama scheme. The MSE_{SDE} is formulated as:

$$MSE_{SDE} = \frac{1}{\mathcal{M}} \sum_{i=1}^{\mathcal{M}} \left(\|S_{i+1} - S_i\|^2 + \|E_{i+1} - E_i\|^2 + \|I_{i+1} - I_i\|^2 + \|R_{i+1} - R_i\|^2 + \|V_{i+1} - V_i\|^2 \right),$$

where $\mathcal{M}$ denotes the total temporal steps. During Monte Carlo sampling, a repository maintains compartment values (S, E, I, R , and V) following each iteration. This repository documents the compartmental evolution across iterations and serves as the foundation for computing the mean solution of the stochastic framework. Upon completion of sampling, the algorithm derives the mean for each compartment from the stored values. This constitutes the stochastic model solution, offering enhanced stability in compartment value estimates by reducing stochastic fluctuations. To assess model performance, the algorithm computes a composite loss function combining data loss (MSE_{SEIRV}) and residual loss ($MSE_{residual}$). The data loss quantifies discrepancies between model predictions (S, E, I, R, V) and empirical observations (S_0, E_0, I_0, R_0, V_0). This is expressed as:

$$MSE_{SEIRV} = \frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \left(\|S - S_i\|^2 + \|E - E_i\|^2 + \|I - I_i\|^2 + \|R - R_i\|^2 + \|V - V_i\|^2 \right),$$

where $\mathcal{N}$ indicates the observation count per compartment, corresponding to collected temporal instances. The residual loss $MSE_{residual}$ represents the mean MSE_{SDE} across all Monte Carlo iterations. This loss quantifies deviations between SDE-based compartment evolution predictions and empirical observations:

$$MSE_{residual} = \frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} MSE_{SDE},$$

where N_{MC} represents Monte Carlo iteration count. The total loss function aggregates data loss (MSE_{SEIRV}) and residual loss ($MSE_{residual}$):

$$Loss = MSE_{SEIRV} + MSE_{residual}.$$

The algorithm uses the Adam optimizer (Torku et al., 2023), a widely adopted optimization technique, to update weights, biases, and dynamic parameters. The Adam optimizer modifies parameters based on the gradients of the loss function. Through iterative parameter updates via the optimizer, the algorithm seeks to minimize the loss function and enhance model prediction accuracy.

In our computational framework, the neural network architecture accepts a singular input parameter, designated as t . The network comprises multiple hidden layers, where inter-nodal connections are characterized by weight matrices $W[i, j]$. Here, i denotes the origin node index, while j represents the destination node index. The hyperbolic tangent activation function is implemented at each node within the hidden layers; that is, $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$. This activation mechanism transforms the input parameter x to a bounded interval spanning -1 to 1 , thereby incorporating nonlinear characteristics into the neural architecture. By using $\tanh$ as the activation function across hidden-layer nodes, the network demonstrates the ability to capture sophisticated patterns and identify intricate data relationships. For the network's output nodes, we use the sigmoid activation function $\sigma(x) = 1/(1 + e^{-x})$ to normalize $S(t), E(t), I(t), R(t)$, and $V(t)$. This activation mechanism transforms input x to values constrained between 0 and 1 , thereby facilitating probability representation or normalized value ranges. The S-PINNs architecture comprises four hidden layers, each containing 64 neurons.

5. The SARS-CoV-2 epidemic in Austria, Switzerland, and Belgium

This section uses real-world SARS-CoV-2 data to illustrate our theoretical results. The data is obtained from Our World in Data (<https://ourworldindata.org/coronavirus>) and covers daily SARS-CoV-2 data from January 10, 2021, to September 30, 2022. Our primary study region is Austria. Before analyzing the data, we preprocess the dataset weekly using a moving-average window.

The first step in preprocessing the reported data is to extract values from the records for different compartments. Using data from Our World in Data, we obtain the number of individuals in compartments I, R , and V based on the known daily new infections, cumulative infections, vaccinated population, and deaths (D). Additionally, based on the World Health Organization's estimate of the SARS-CoV-2 incubation period, we derive the number of individuals in the incubation period (E). Thus, $S = N - I - E - R - V - D$, and the other parameters of the model are shown in Table 1.

To evaluate the accuracy of the fitted models, we utilize two statistical metrics: the mean absolute percentage error (MAPE) and the normalized root mean square error (NRMSE) below:

Table 1
Parameters and description.

Parameter	Description	Value (Source)
Λ	Birth rate	0.0085 (Goisis et al., 2021)
β	Transmission rate	0.1–0.9 (S-PINNs)
μ_S, μ_R, μ_V	Death rates for the S, R, V	0.0103 (Pilz et al., 2025)
μ_E	Death rate for the E	0.0031 (Perone, 2025)
μ_I	Death rate for the I	0.0031 (Perone, 2025)
α_E	Conversion rate from the exposed to the infected	0.1781 (He, Tang, & Xiao, 2023)
γ_I	Recovery rate	0.1 (He, Tang, Xiao, et al., 2023)
ϵ	Vaccination rate	Time-varying (Mathieu et al., 2024)
η	Intensity of government control	0.1–0.9 (S-PINNs)
$\mathcal{D}$	Public dynamic behavior	0.1–0.9 (S-PINNs)
$\mathcal{K}$	Degree of public response	$\mathbb{R}^+$ (S-PINNs)

$$\text{MAPE}(Y_i, \hat{Y}_i) = \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right| \times 100\%, \quad \text{NRMSE}(Y_i, \hat{Y}_i) = \frac{1}{\bar{Y}} \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2},$$

where Y_i represents the observed values, $\hat{Y}_i$ denotes the predicted or fitted values, n is the total number of observations, and $\bar{Y}$ is the mean of the observed values. The MAPE is a percentage-based measure of relative error, and NRMSE is a normalized measure of absolute deviation.

To comprehensively evaluate the performance and robustness of the S-PINNs model, we design a series of comparative experiments. Initially, we conduct fitting experiments using real SARS-CoV-2 transmission data from Austria, comparing S-PINNs with conventional PINNs models under different noise conditions. Specifically, we artificially introduce Gaussian white noise at three distinct levels (5%, 10%, and 15%) into the original dataset to simulate potential measurement errors, statistical uncertainties, and other stochastic factors that may arise during actual data collection. Using this progressive noise-level configuration, we systematically investigate the fitting performance and stability of both models under noisy conditions. The 5% noise level represents relatively ideal data quality conditions; the 10% noise level simulates moderate data uncertainty; and the 15% noise level tests the models' robustness and fault tolerance under extreme conditions. This multi-tier noise-testing approach not only facilitates validating the S-PINNs model's superiority over conventional PINNs but also provides crucial theoretical foundations and performance benchmarks for model reliability in practical applications.

Figs. 5 and 6 compare the performance of PINNs and S-PINNs on noisy data at 5% and 15% noise levels, respectively. As noise intensity increases, the S-PINNs exhibit greater adaptive fluctuations, making them markedly superior to PINNs. Each figure consists of two sets of comparative curves, with each set representing the fitting results of PINNs and S-PINNs, respectively. These noisy data points are generated by artificially introducing 5% and 15% noise levels into real SARS-CoV-2 epidemic data, to simulate the effects of data uncertainty and measurement errors in practical applications.

Through comparative analysis, it is clear that the S-PINNs demonstrate superior robustness and fitting accuracy in handling noisy environments. When the noise level increases from 5% to 15%, the performance gap between the two models becomes increasingly evident. S-PINNs not only maintain high fitting quality but also exhibit strong resistance to data contamination. The black dashed lines typically represent the fitting effects of S-PINNs, while the blue dashed lines represent the fitting results of PINNs. This intuitive comparison clearly illustrates the superior noise tolerance and adaptability of the S-PINNs approach in the face of inevitable data uncertainty in real-world applications.

Table 2 presents a comparative analysis between PINNs and S-PINNs using NRMSE and MAPE. For each noise condition, two columns demonstrate the performance of the compared models: the deterministic approach (PINNs) and the stochastic approach (S-PINNs). The results reveal that S-PINNs consistently outperform PINNs across all examined noise levels, achieving higher accuracy in both NRMSE and MAPE. These findings underscore the remarkable ability of S-PINNs to effectively capture the dynamic behaviors and inherent uncertainties in noisy datasets. By incorporating stochastic modeling methodologies, S-PINNs achieve significantly improved prediction accuracy and demonstrate superior data-fitting capabilities compared to their deterministic PINN counterparts. The fundamental advantage of S-PINNs lies in their ability to effectively model and quantify the intrinsic variability and uncertainty in the data, thereby establishing them as a more robust and reliable approach for modeling and forecasting complex systems subject to noise-induced perturbations. This enhanced performance particularly manifests in the models' capacity to handle uncertainty quantification, making S-PINNs an optimal choice for applications where data corruption and stochastic variations are prevalent challenges in system identification and predictive modeling tasks. To further validate the effectiveness of our models, we selected 60 days of data from Austria, Switzerland, and Belgium for experimentation, and split the dataset into training and test sets at an 8:2 ratio. Based on this experimental setup, we conducted a comparative analysis of predictive performance between our PINNs and S-PINNs models and two baseline models: the Long Short-Term Memory (LSTM) network and the Logistic model. LSTM is a specialized variant of Recurrent Neural Networks (RNNs) designed to capture and predict long-term dependencies in time series and sequential data. On the other hand, the Logistic model constitutes a classical epidemiological framework

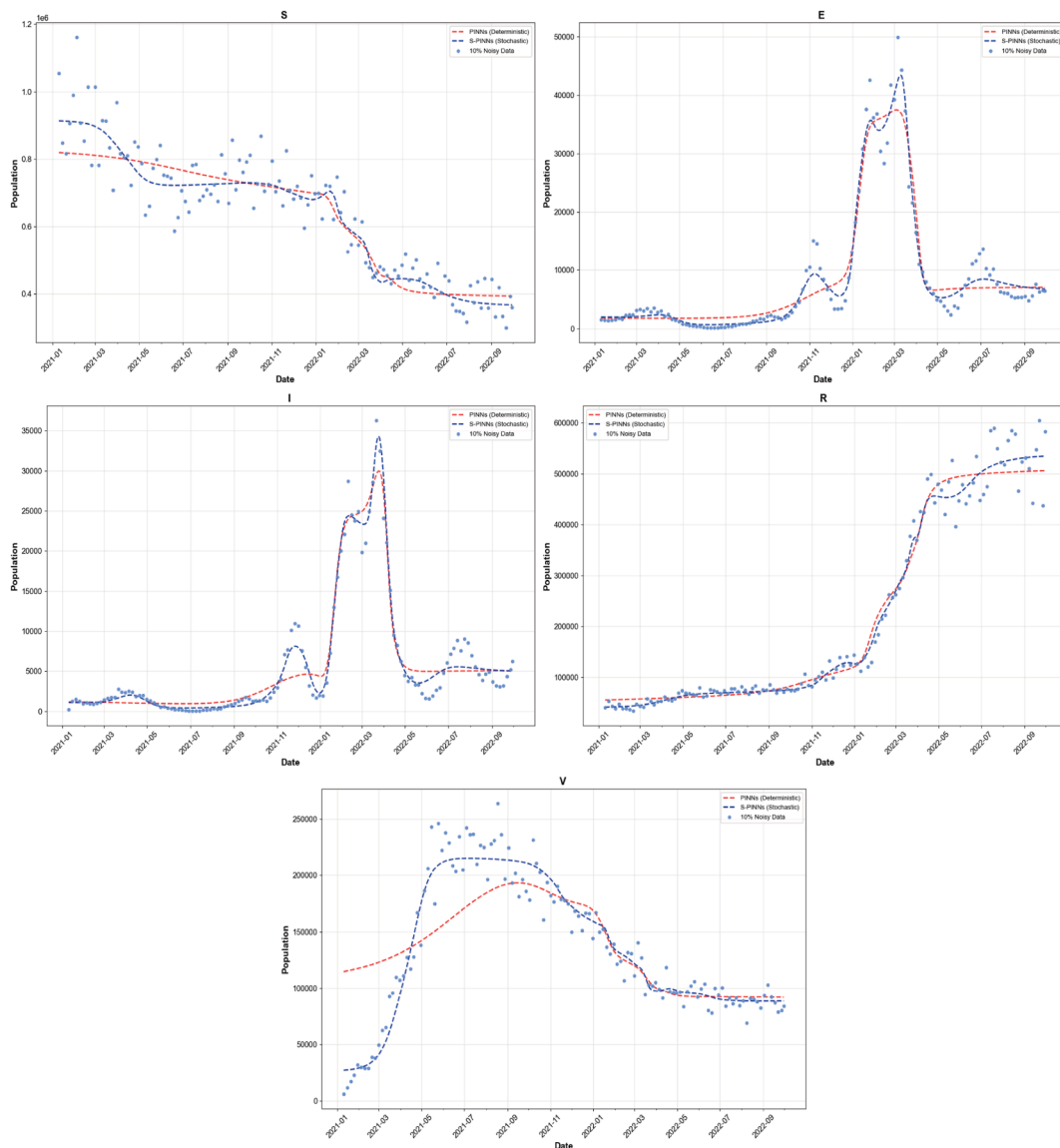


Fig. 5. Comparison of PINNs and S-PINNs performance on data with 5% noise. The blue scatter points represent the noisy data, the blue dashed line shows the fitting result of S-PINNs, and the red dashed line shows the fitting result of PINNs.

that assumes disease transmission follows an S-shaped curve pattern, characterized by slow initial growth, rapid intermediate expansion, and eventual saturation. This model describes the dynamic processes of population growth or disease transmission through differential equations, offering the advantages of structural simplicity, interpretability, and a parsimonious parameter set.

Through this multi-model comparative experimental design, we can systematically evaluate the predictive capabilities and applicability of various modeling approaches under identical data conditions.

To comprehensively evaluate the robustness and predictive performance of different models under noisy environments, we designed systematic comparative experiments. Specifically, we conducted a comparative analysis at noise levels of 5%, 10%, and 15%. These noise levels represent different data quality conditions, ranging from mild to severe, and can adequately reflect various degrees of data contamination encountered in practical applications.

Fig. 7 illustrates the comparison between observed values and predictions across three different noise level datasets for the S-PINNs (red) and other models (PINNs in blue, Logistic model in black, LSTM in purple). Overall, all four data-driven methods can effectively reproduce case growth trends; however, as noise levels increase, the performance differences between models become more pronounced.

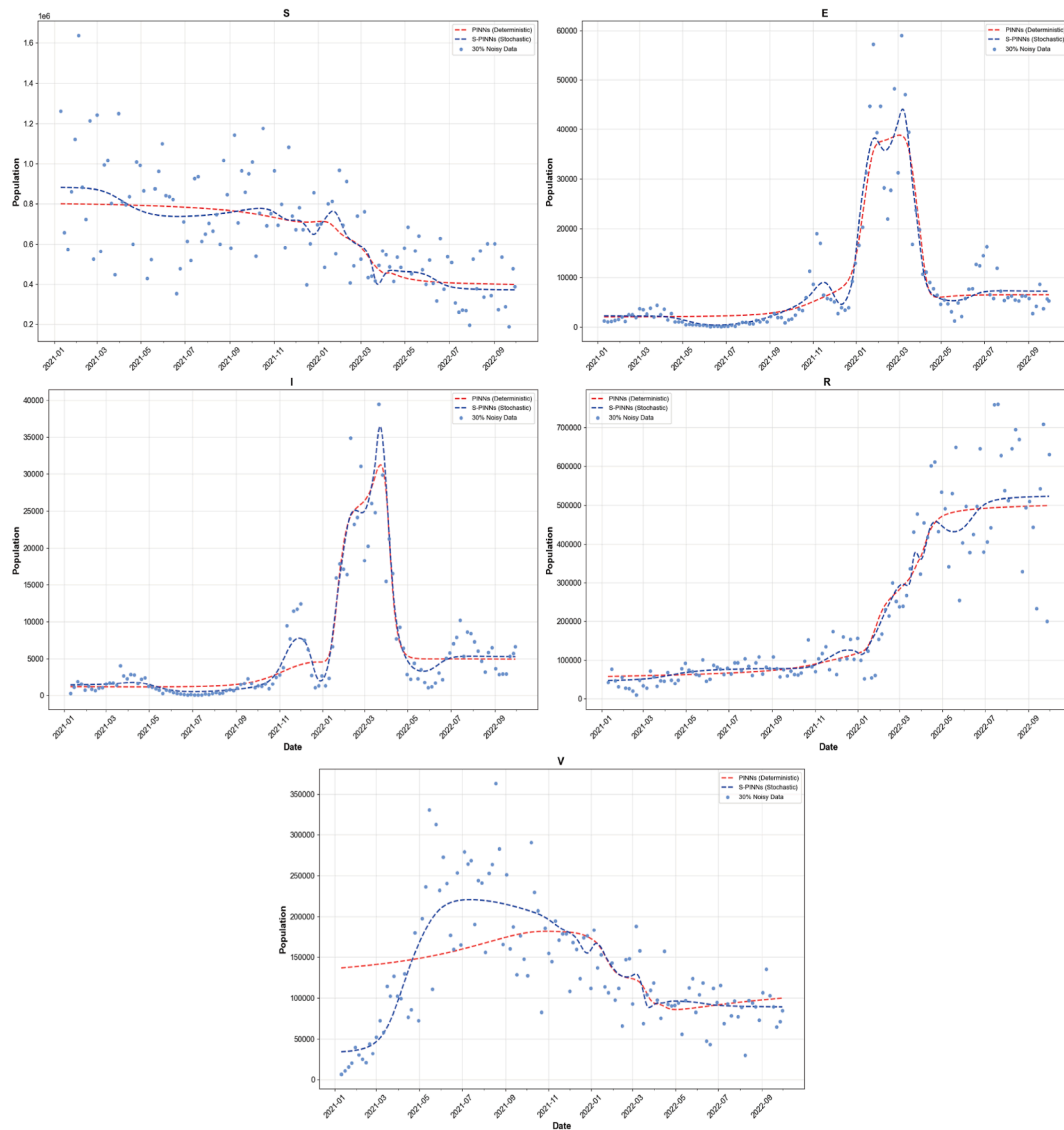


Fig. 6. Comparison of PINNs and S-PINNs performance on data with 15% noise. The blue scatter points represent the noisy data, the blue dashed line shows the fitting result of S-PINNs, and the red dashed line shows the fitting result of PINNs.

Under low-noise conditions ($\sigma = 0.05$), all methods exhibit relatively good fit performance. The Logistic model exhibits typical S-shaped curve characteristics; the PINNs can also track data trends reasonably well; the LSTM model performs stably; and the S-PINNs similarly accurately track the data's developmental trajectory. At this noise level, the differences among the four models are relatively small.

As the noise level increases to a moderate degree ($\sigma = 0.1$), differences between models begin to emerge. The Logistic model maintains its inherent smoothness; the PINNs begin to show some sensitivity to noise but remain relatively stable. The LSTM model exhibits more pronounced fluctuation features, while the S-PINNs model demonstrates excellent balancing, capturing main trends while responding moderately to local variations in the data.

Under high noise conditions ($\sigma = 0.15$), the performance differences between methods become highly significant. The Logistic model's prediction curve remains relatively smooth. Clearly, it deviates from the observed data; the PINNs attempt to fit the data but exhibit instability under extreme noise conditions, and the LSTM model shows violent fluctuations and excessive oscillations. Our S-PINNs demonstrate outstanding robustness and adaptability across all noise levels, maintaining overall trend accuracy while reasonably responding to data uncertainties. This enables it to avoid the over-smoothing and over-fluctuation problems of other models, thereby demonstrating greater practical value.

From Table 3, we can obtain that, in the Austrian dataset, the S-PINNs model exhibits superior performance across all noise levels, with NRMSE increasing progressively from 4.36 at 5% noise to 20.86 at 15% noise, while MAPE rises from 23.98

Table 2
Performance comparison of PINNs and S-PINNs under different noise levels.

Noise Level	Model	NRMSE	MAPE
5%	PINNs	3.00	51.37
	S-PINNs	1.64	17.91
10%	PINNs	3.56	58.32
	S-PINNs	2.81	24.03
15%	PINNs	6.17	88.43
	S-PINNs	5.88	44.86

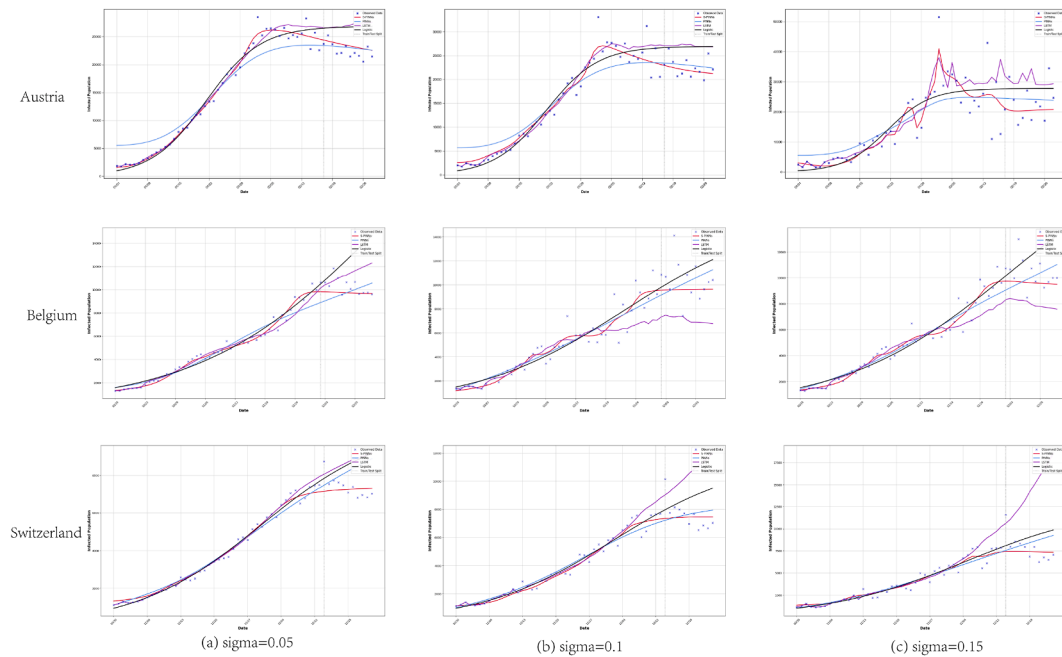


Fig. 7. Comparison of prediction performance among S-PINNs, PINNs, LSTM, and Logistic models in Austria, Switzerland, and Belgium under varying noise levels: (a) 5% noise level ($\sigma = 0.05$); (b) 10% noise level ($\sigma = 0.1$); (c) 15% noise level ($\sigma = 0.15$).

to 36.65, demonstrating stable robustness. In contrast, PINNs shows NRMSE escalating from 15.17 to 33.91, yet MAPE decreases from 77.18 to 56.75, indicating improved relative error in high-noise environments. LSTM displays slight fluctuations in NRMSE from 21.72 to 26.47, with MAPE declining markedly from 106.71 to 45.07. The Logistic model maintains relative stability in NRMSE from 18.83 to 21.48, while MAPE drops from 93.47 to 37.52. In Belgium, S-PINNs similarly leads, with NRMSE ranging from 5.14 to 11.95 and MAPE decreasing from 33.19 to 25.94, reflecting adaptability as noise intensifies. PINNs exhibits NRMSE from 11.40 to 14.51 and a substantial MAPE reduction from 58.14 to 27.28. LSTM's NRMSE rises from 9.93 to 30.29, with MAPE remaining largely stable from 52.21 to 54.08. Logistic shows NRMSE decreasing from 23.28 to 16.34 and MAPE sharply falling from 125.35 to 29.07. In Switzerland, S-PINNs sustains its advantage, with NRMSE from 5.99 to 12.08 and MAPE declining from 30.59 to 27.17. PINNs has NRMSE from 14.73 to 20.49 and MAPE dropping from 64.47 to 36.70. LSTM's NRMSE surges from 19.78 to 93.43, with MAPE increasing from 79.74 to 140.34, highlighting sensitivity to high noise. Logistic exhibits NRMSE from 18.61 to 25.35 and MAPE decreasing from 76.42 to 42.14.

Overall, S-PINNs performs exceptionally across the datasets from all three countries, accurately fitting trends under low noise while maintaining low error levels in high-noise conditions, underscoring its advantages in integrating physical constraints and noise-handling mechanisms. As a foundational physics-informed neural network, PINNs excels in capturing physical constraints but remains sensitive to noise; the observed MAPE decline under high noise may stem from the model shifting toward more generalized smooth predictions, mitigating overfitting prevalent in low-noise scenarios. LSTM, as a recurrent network, is adept at modeling long-term dependencies but prone to noise-induced fluctuations; in cases like Austria, the MAPE reduction could result from high noise compelling more robust pattern learning, whereas deterioration in Switzerland reveals inherent instability. The Logistic model, as a classical parametric framework, features simplicity and smoothness, often outperforming complex models under high noise; its widespread MAPE decline can be attributed to innate resistance to noise, eschewing local fluctuations in favor of aligning with overarching S-shaped trends. According to the literature, such as in (Yin et al., 2023), MAPE as a scale-sensitive metric tends to amplify relative errors when data

Table 3

Performance comparison of S-PINNs, PINNs, LSTM, and Logistic under different noise levels and regions.

Region	Noise Level	Model	NRMSE	MAPE
Austria	5%	S-PINNs	4.36	23.98
		PINNs	15.17	77.18
		LSTM	21.72	106.71
		Logistic	18.83	93.47
	10%	S-PINNs	8.33	31.71
		PINNs	18.71	67.40
		LSTM	26.17	90.91
		Logistic	19.46	69.80
	15%	S-PINNs	20.86	36.65
		PINNs	33.91	56.75
		LSTM	26.47	45.07
		Logistic	21.48	37.52
Belgium	5%	S-PINNs	5.14	33.19
		PINNs	11.40	58.14
		LSTM	9.93	52.21
		Logistic	23.28	125.35
	10%	S-PINNs	8.81	28.29
		PINNs	11.84	32.23
		LSTM	22.08	56.94
		Logistic	18.54	48.46
	15%	S-PINNs	11.95	25.94
		PINNs	14.51	27.28
		LSTM	30.29	54.08
		Logistic	16.34	29.07
Switzerland	5%	S-PINNs	5.99	30.59
		PINNs	14.73	64.47
		LSTM	19.78	79.74
		Logistic	18.61	76.42
	10%	S-PINNs	9.32	27.84
		PINNs	11.53	32.13
		LSTM	43.61	94.60
		Logistic	21.79	50.32
	15%	S-PINNs	12.08	27.17
		PINNs	20.49	36.70
		LSTM	93.43	140.34
		Logistic	25.35	42.14

include small values or low noise, whereas increased noise may homogenize or enlarge actual value distributions, thereby reducing MAPE values—a common phenomenon in time-series forecasting, particularly when models like Logistic favor underestimation. Similarly, [Vivas et al., \(2020\)](#) discussed MAPE reductions when accounting for sources of variation, implying that noise as a variability factor can enhance relative accuracy through averaging effects.

6. Conclusion

This paper proposes a novel stochastic SEIRV model, based on the S-PINNs algorithm, to study the dynamics of infectious disease transmission. The research introduces stochastic perturbations to deterministic models, providing a more realistic reflection of the uncertainty and complexity of infectious disease transmission in the real world.

We constructed a stochastic SEIRV system that accounts for environmental noise and random fluctuations, dividing the population into five compartments with Brownian perturbations. By constructing appropriate Lyapunov functions and applying Itô's formula, we rigorously proved the global positivity of solutions, ensuring the biological reasonableness. Applying ergodic theory, we analyzed the long-term behavioral characteristics of the stochastic system, proved the ergodic properties of the disease, and established criteria for disease persistence and extinction based on the threshold value R_0^* . Moreover, we provided an analytical expression for the stationary distribution. Numerical simulations employed the high-precision Milstein higher-order method to verify theoretical predictions through carefully designed parametric experiments. The results demonstrate that by appropriately adjusting government control intensity and public response parameters, disease transmission dynamics can be effectively controlled, resulting in significantly reduced infection peaks and a shorter epidemic duration. The numerical experiments fully illustrate the crucial role of vaccination strategies in epidemic prevention and control, providing a quantitative basis for formulating optimal vaccine allocation strategies.

To evaluate the actual application performance of the model, we proposed an innovative data-physics fusion hybrid algorithm, called S-PINNs, which represent an essential extension based on classical PINNs, with the core innovation being the explicit introduction of stochastic noise terms in the loss function, enabling the network to more reasonably learn and capture the noise characteristics and uncertainties commonly present in real epidemic data. By embedding the physical

constraints of stochastic differential equations into the loss function, we achieved organic integration of data-driven learning and physical law constraints. The application to actual SARS-CoV-2 epidemic data from Austria, Switzerland, and Belgium validated the theoretical framework. By comparing the predictive performance of PINNs, LSTMs, and classical Logistic growth models across different noise levels, our model demonstrated comparable or even superior predictive accuracy, particularly excelling at predicting noisy data. This suggests that the proposed method possesses strong generalization capabilities and can be applied to other infectious diseases in complex environments.

In conclusion, the stochastic SEIRV model, combined with the S-PINNs-based hybrid method proposed in this study, provides a powerful and innovative framework for research on infectious disease dynamics. This framework not only has a rigorous mathematical foundation theoretically but also demonstrates excellent predictive performance in practical applications, offering more precise and reliable scientific support for the formulation of public health policies. Despite significant progress, several challenges remain to be addressed in future work, considering the model's assumptions and constraints. For instance, omitting an explicit R compartment could lead to a systematic overestimation of the susceptible population size, thereby underestimating the vaccination coverage threshold required for herd immunity or overestimating the sustained infection levels at the endemic equilibrium. Similarly, if infection exerts a significant adverse effect on fertility that is not accounted for in the model, it may underestimate the disease's profound impacts on population structure and long-term transmission dynamics. To assess the robustness of these assumptions and enhance the model's real-world explanatory power, future extensions could include: adopting a more comprehensive SEIRV framework; incorporating waning immunity mechanisms; explicitly modeling fertility reduction induced by infection, complemented by thorough sensitivity analyses to systematically quantify the specific effects of these factors on key predictive indicators. The emergence of viral variants may alter transmission dynamics, vaccine efficacy may fluctuate over time or with viral mutations, and the model has not fully incorporated environmental and behavioral factors. These limitations necessitate further investigation through the integration of multi-strain competition submodels or the incorporation of real-time genomic sequencing data to enhance the model's robustness and applicability.

CRediT authorship contribution statement

Linfeng Xie: Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Jikai Yang:** Writing – review & editing, Software, Methodology. **Zhiming Li:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data supporting the findings of this study are openly available at <https://ourworldindata.org/coronavirus>.

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